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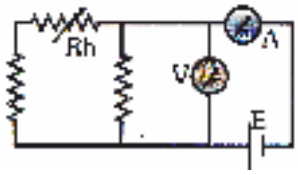
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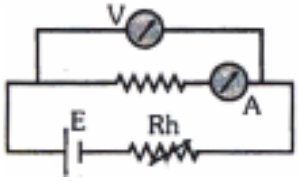
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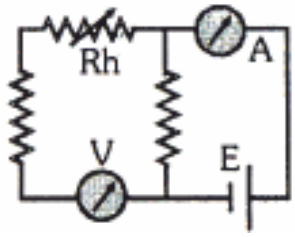
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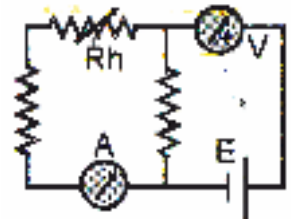
PHYSICS

1) Express which of the following setups can be used to verify Ohm's law :

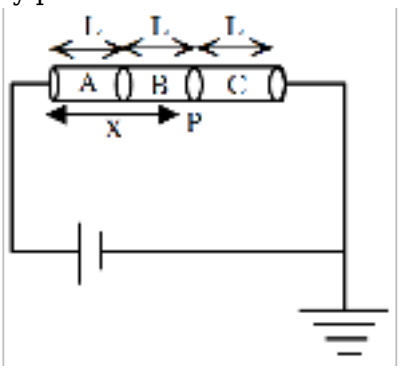
(1) 

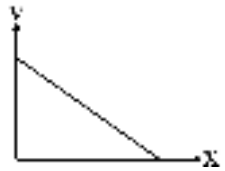
(2) 

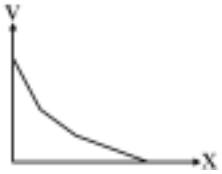
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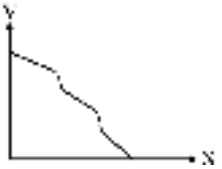
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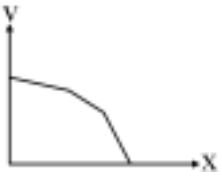
2) Three cylindrical rods A, B and C of same length and same cross-sectional area are joined in series and connected to an ideal cell as shown in figure. The resistivities of rods are in ratio of 1:2:3 respectively. The veriation of potential at any point P at a distance x from the left end of combined rods is most appropriatly shown in figure :



(1) 

(2) 

(3) 

(4) 

3) Power generated across a uniform wire connected across a supply is H. If the wire is cut into n equal parts and all the parts are connected in parallel across the same supply, the total power generated in the wire is

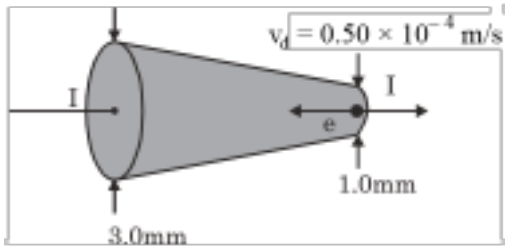
(1) $\frac{H}{n^2}$

(2) n^2H

(3) nH

(4) $\frac{H}{n}$

4) What is the electron drift speed at the 3 mm diameter end (the left end) of the wire in figure.



(1) 0.055×10^{-4} m/s

(2) 4.5×10^{-4} m/s

(3) 0.055×10^{-3} m/s

(4) 4.5×10^{-3} m/s

5) If n,e,τ,m, are representing electron density, charge, relaxation time and mass of an electron respectively then the resistance of wire of length and cross sectional area A is given by :-

(1) $\frac{m\ell}{ne^2\tau A}$

(2) $\frac{2mA}{ne^2\tau}$

(3) $ne^2\tau A$

(4) $\frac{ne^2\tau A}{2m}$

6) In experiment of measurement of internal resistance of a cell the reading of ammeter is 2A when load resistance (R_1) is 3Ω. If load resistance is 6 Ω the reading of ammeter is 1.25A. What will be reading of ammeter when load is 8Ω?

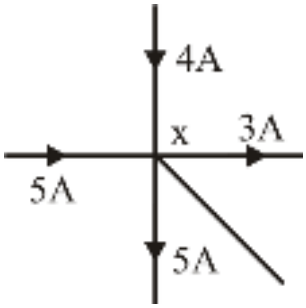
(1) 1A

(2) $\frac{8}{9}$A

(3) $\frac{3}{4}$A

(4) $\frac{4}{5}$A

7) Five conductors are meeting at a point x as shown in the figure. What is the value of current in fifth conductor :-



(1) 3 A away from x

(2) 1 A away from x

(3) 4 A away from x

(4) 1 A towards x

8) Three copper wires A,B,C of length ratio 2 : 3 : 4 and mass ratio 1 : 3 : 2 are connected in series then find ratio of potential drop across wires :-

(1) 3 : 4 : 8

(2) 4 : 3 : 8

(3) 2 : 3 : 4

(4) 1 : 1 : 5

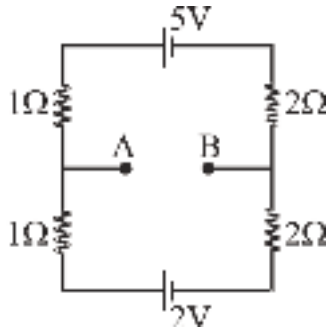
9) A wire of resistance R is bent in form of a circle. Find net resistance between two points A and B on circumference of this circle where arc AB substandes π/3 angle at centre :-

(1) $\frac{5}{36}$R

(2) $\frac{7}{36}$R

(3) $\frac{36}{5}$R

(4) $\frac{36}{7}$R

10) 

Find out potential difference between point A and B.

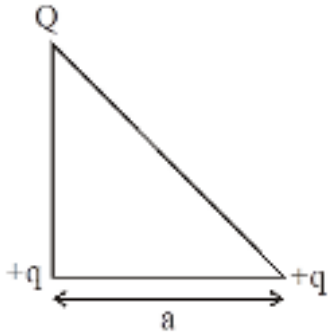
(1) 3.5 V

(2) 3 V

(3) 4 V

(4) 5 V

11) Three charges Q, + q and + q are placed at the vertices of a right – angled isosceles triangle as shown. The net electrostatic energy of the configuration is zero if Q is equal to :-



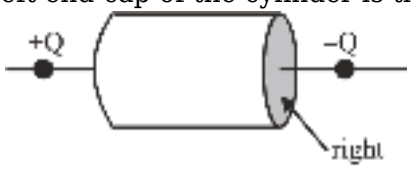
(1) $\frac{-q}{1+\sqrt{2}}$

(2) $\frac{-2q}{2+\sqrt{2}}$

(3) –2q

(4) +q

12) Positive and negative charges of equal magnitude lie along the symmetry axis of a cylinder. The distance from the positive charge to the left end-cap of the cylinder is the same as the distance from the negative charge to the right end cap.



The sign of the flux through the right end-cap of the cylinder is :

(1) Positive

(2) Negative

(3) There is no flux through the right end-cap.

(4) None of these

13) The electric field is 100 V/m, at a distance of 20 cm from the centre of a dielectric sphere of

radius 10 cm. Then E at 3 cm distance from the centre of sphere is :–

(1) 100 V/m
(2) 125 V/m
(3) 120 V/m
(4) zero

14) A charged soap bubble of radius r is in equilibrium with air pressure inside and outside equal. If T is the surface tension of soap solution, then potential of the soap bubble is -

(1) $\sqrt{8T\varepsilon_0/r}$
(2) $\sqrt{8Tr/\varepsilon_0}$
(3) $\sqrt{8Tr\varepsilon_0}$
(4) none of these

15) ***Assertion (A) :*** Two concentric conducting spherical shells are charged. The charge on the outer shell is varied keeping the charge on inner shell constant, as a result the electric potential difference between the two shells does not change.
Reason (R) : In the situation of Assertion, the electric field in the region between both the shells depends only on the charge existing on the inner shell. :-

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
(2) **(A)** is correct but **(R)** is not correct
(3) **(A)** is incorrect but **(R)** is correct
(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

16) Consider system X of 3 bodies A, B and C. If charge on C is 2C and charge on system X is 10C then charges of A and B can be :-

(1) A = 2C, B = 4C
(2) A = 3C, B = 6C
(3) A = –4C, B = –4C
(4) A = –4C, B = 12C

17) Two similar spheres having +q and –q charge are kept at a certain distance. F force acts between them. If in the middle of two spheres, another similar sphere having +q charge is kept, then it experience a force in magnitude and direction as :-

(1) Zero having no direction
(2) 8F towards +q charge
(3) 8F towards –q charge
(4) 4F towards +q charge

18) Point charges +4q, –q and +4q are kept on the x-axis at point x = 0, x = a and x = 2a respectively, then :-

(1) Only –q is in stable equilibrium

(2) None of the charges are in equilibrium

(3) All the charge are in unstable equilibrium

(4) All the charges are in stable equilibrium

19) Surface charge density of a thin disc having radius R varies with distance r from centre as $\sigma = \frac{\sigma_0 r^2}{R^2}$, ($r \neq 0$) then total charge of disc is :-

(1) $2\pi\sigma_0 R^2$

(2) $\sqrt{2}\pi\sigma_0 R^2$

(3) $2\sqrt{2}\pi\sigma_0 R^2$

(4) $\frac{\pi\sigma_0 R^2}{2}$

20) A charged spherical shell of diameter 4 cm has a charge density of 10^{-4} coulomb/cm^2. The work done in joule when a charge of 40 nano-coulombs is moved from infinity to a point which is at a distance of 2 cm from the surface of the sphere is :-

(1) 14.4 π

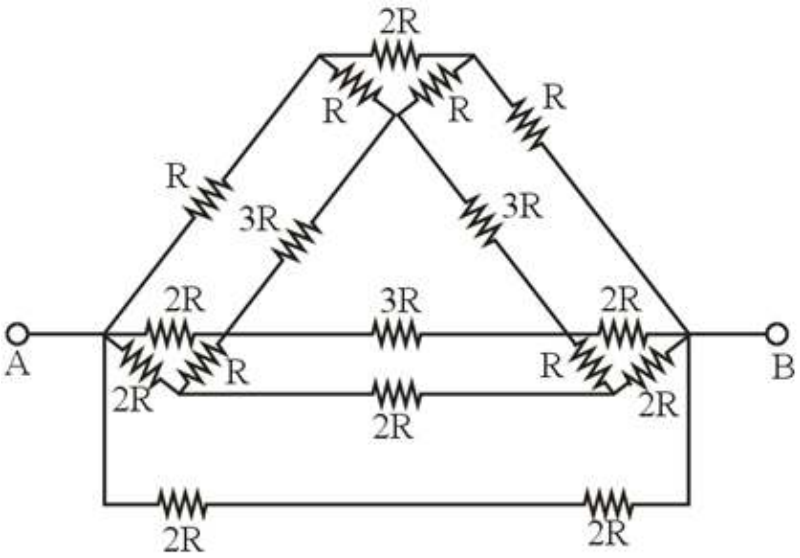
(2) 28.8 π

(3) 144 π

(4) 288 π

21)

Find equivalent resistance between A & B.



(1) $\frac{R}{11}$

(2) $\frac{12R}{11}$

(3) R

(4) 4R

22) The temperature coefficient of resistivity of a material is 4×10^{-4}/K. When the temperature of material is increased by 50°C, its resistivity increases by 2×10^{-8} Ωm. The initial resistivity of material (in ohm-metre) is :

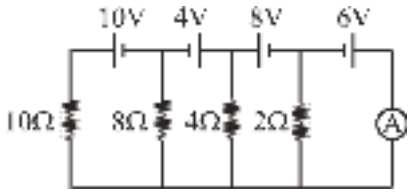
(1) 50×10^{-8}

(2) 90×10^{-8}

(3) 100×10^{-8}

(4) 200×10^{-8}

23) Find the reading of ideal ammeter :-



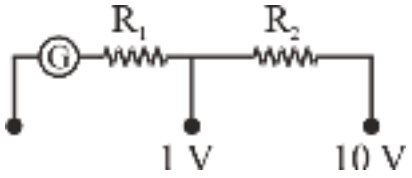
(1) 0.8 A

(2) 0.25 A

(3) 1.95 A

(4) 0.05 A

24) A galvanometer has an internal resistance of 50 Ω and current required for full scale deflection is 1mA find the series resistance (R_1 & R_2) required to use it as a voltmeter with range as indicated in figure :-



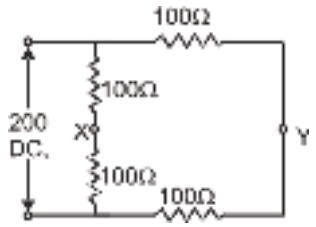
(1) 950 Ω, 9 kΩ

(2) 950 Ω, 10 kΩ

(3) 9950 Ω, 90 kΩ

(4) None of these

25) $V_x - V_y$ will be ?

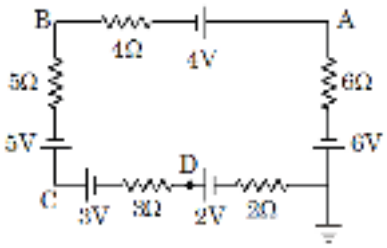


(1) 10

(2) 50

(3) 100

(4) 0



26) In the given figure, all the batteries are ideal.

	List-I		List-II
(P)	Potential of point A is	(1)	–5V
(Q)	Potential of point B is	(2)	1V
(R)	Potential of point C is	(3)	–3V
(S)	Potential of point D is	(4)	2.5V

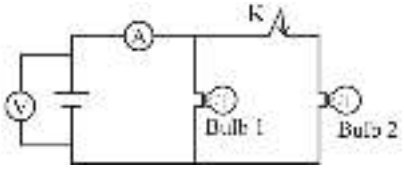
(1) P→1; Q→2; R→3; S→4

(2) P→4; Q→1; R→2; S→3

(3) P→2; Q→3; R→4; S→1

(4) P→3; Q→1; R→4; S→2

27) In the given circuit, on pressing the key K, if bulb are identical, then



(I) Bulb-1 glows brighter than bulb-2

(II) Bulb-2 glows brighter than bulb-1

(III) Reading of ammeter become twice

(IV) Reading of voltmeter remains same

(1) I & III

(2) II & IV

(3) III & IV

(4) I, III & IV

28) Two identical charged balls each of mass m are suspended from a common point. Length of each string is □. In equilibrium distance between balls is r then what will be tension force in string :-

(1) $\frac{mg}{\sqrt{\ell^2 - r^2}}$

(2) $\frac{mg\ell}{\sqrt{\ell^2 - \frac{r^2}{4}}}$

(3) mg

(4) $\frac{mg}{\ell}$

29) Two point charges –q and +q are kept at points (0, 0, –a) and (0, 0, a). Find the electric potential at point (0, 0, z) where z > a :-

(1) $\frac{qa}{4\pi\varepsilon_0 z^2}$

(2) $\frac{q}{4\pi\varepsilon_0 a}$

(3) $\frac{2qa}{4\pi\varepsilon_0(z^2-a^2)}$

(4) $\frac{2qa}{4\pi\varepsilon_0(z^2+a^2)}$

30) Lines of a electrostatic field can't have sudden break in charge free region, because :-

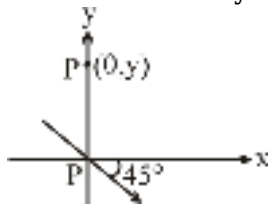
(1) It can't formed closed loop

(2) It is imaginary line

(3) Gauss's theorem will not be true

(4) These line can't exist in charge free region

31) A short electric dipole is placed at origin of coordinate system as shown in figure, find the electric field at point P(0, y) $\left(k=\frac{1}{4\pi\in_0}\right)$:



(1) $\left(\sqrt{\frac{5}{2}}\right)\frac{kP}{y^3}$

(2) $\sqrt{5}\frac{kP}{y^3}$

(3) $\left(\sqrt{\frac{3}{2}}\right)\frac{kP}{y^3}$

(4) $\sqrt{3}\frac{kP}{y^3}$

32) Amount of work done in moving a charge q_0 from centre of ring to an axial point at $x = \sqrt{8}a$ will be, if charge given to ring is Q and 'a' being the radius:-

(1) $\frac{Qq_0}{6\pi\varepsilon_0 a}$

(2) $-\frac{Qq_0}{6\pi\varepsilon_0 a}$

(3) Zero

(4) $\frac{Qq_0}{4\pi\varepsilon_0 a}$

33)

The equipotential surfaces due to a point charge q, whose potentials differ by a constant amount

(say 1.0 volt) are :

(1) equally spaced
(2) unequally spaced, separation increasing as one goes away from q
(3) unequally spaced, separation decreasing as one goes away from q
(4) none of the above

34) Two large conducting plates are placed parallel to each other with a separation of d between them. An electron starting from rest near one of the plates reaches the other plate in time t. If e is the charge on the electron and m is its mass, then the surface charge density on the inner surface is :

(1) $\frac{dme}{4\pi\varepsilon_0 t^2}$
(2) $\frac{dm\varepsilon_0}{4\pi et^2}$
(3) $\frac{2dm\varepsilon_0}{et^2}$
(4) none of these

35) An electric dipole of length 1 cm is placed with the axis making an angle of 30° to an electric field of strength 10^4 NC^{-1}. If it experiences a torque of $10\sqrt{2}$ Nm, the potential energy of the dipole is :

(1) –0.245 J
(2) 0.245 J
(3) 24.5 J
(4) –24.5 J

36) A meter bridge is used with a standard 12 Ω resistor to determine the resistance of a coil of 1 meter long unknown wire. The balance point is obtained at 60 cm along the meter bridge wire. If the area of cross-section of unknown wire is 0.2×10^{-6} m^2, the resistivity of this unknown wire is (in Ω – m) :

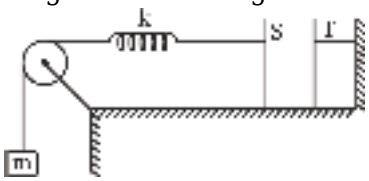
(1) 1.6×10^{-6}
(2) 4×10^{-6}
(3) 8×10^{-5}
(4) 12×10^{-6}

37) During meter bridge experiment the null point was obtained at distance of 40cm from terminal A (begining of scale) when known Resistance R is connected to left gap near A. When a resistance of 10Ω is connected parallel to unknown resistance X in right gap near to terminal B. It was found that null point shifted to 60cm. The value of unknown resistance X is :

(1) 7.5 Ω
(2) 9.0 Ω
(3) 10.5 Ω

(4) 12.5 Ω

38) The plates S and T of an uncharged parallel plate capacitor are connected across a battery. The battery is then disconnected and the charged plates are now connected in a system as shown in the figure. The system shown is in equilibrium. All the strings are insulating and massless. The magnitude of charge on one of the capacitor plates is : [Area of plates = A]



(1) $\sqrt{2mgA\epsilon_0}$

(2) $\sqrt{\frac{4mgA\epsilon_0}{k}}$

(3) $\sqrt{mgA\epsilon_0}$

(4) $\frac{\sqrt{2mgA\epsilon_0}}{k}$

39) A parallel plate capacitor has a capacitance of 2 μF. A dielectric slab of constant K = 5 is inserted between the plates and the capacitor is charged to 100 V and isolated. If the dielectric slab is now removed then the new potential difference will be :-

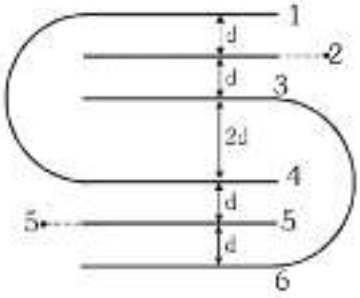
(1) 250 V

(2) 500 V

(3) 100 V

(4) 20 V

40) There are six parallel plates of equal area A are arranged as shown in figure (d<<A). The equivalent capacitance between points 2 and 5, is $\alpha\frac{\epsilon_0 A}{d}$. Then find the value of α.



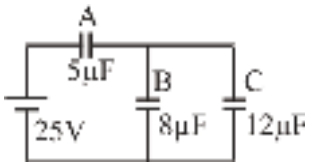
(1) 1

(2) 2

(3) 3

(4) 4

41) Three capacitor A, B and C are connected to a battery of 25 volt as shown in the figure. The ratio

of charges on capacitors A, B and C will be :- 

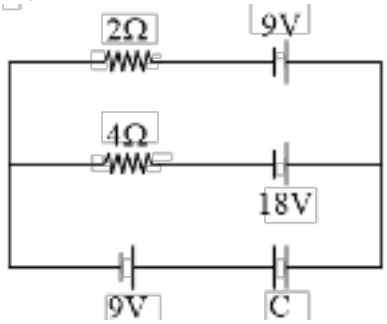
(1) 5 : 2 : 3

(2) 5 : 3 : 2

(3) 2 : 5 : 3

(4) 2 : 3 : 5

42) In the circuit shown, calculate the potential drop (in volt) across the capacitor.



(1) 9 V

(2) 12 V

(3) 3 V

(4) 0 V

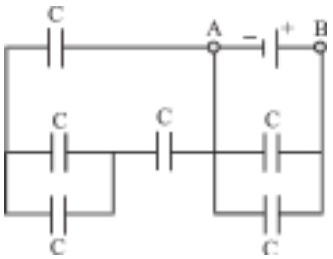
43) In a charged capacitor, the energy store:

(1) At positive charge plate

(2) At both the positive and negative charge plates

(3) In field between the plates

(4) Around the edge of the capacitor plates

44) Find equivalent capacitance between A and B:- 

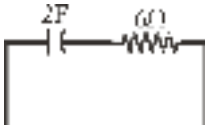
(1) 3 C

(2) 5 C

(3) 4 C

(4) 2 C

45) The condenser shown in the circuit is charged to 5V and left in the circuit, in 12s the charge on

the condenser will become (Where e = 2.718) 

(1) $\frac{10}{e}C$

(2) $\frac{e}{10}C$

(3) $\frac{5}{e}C$

(4) $\frac{e}{5}C$

CHEMISTRY

1) The enthalpies of neutralization of a weak base AOH and a strong base BOH by HCl are –12250 cal/mol and –13000 cal/mol respectively. When one mole of HCl is added to a solution containing 1 mole of AOH and 1 mole of BOH, the enthalpy change was –12500 cal/mol. In what mole ratio the acid neutralise base AOH and BOH ?

(1) 2 : 1

(2) 2 : 3

(3) 1 : 2

(4) 3 : 2

2) **Statement-I :** In the preparation of As_2S_3 sol by passing H_2S gas through a solution of As_2O_3, the resulting colloidal particles carry a negative charge.
Statement-II : The negative charge on As_2S_3 colloidal particles is due to the preferential adsorption of sulphide ions (S^{-2}) from the ionisation of H_2S.

(1) Statement I is incorrect but Statement II is correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I and Statement II both are correct.

(4) Statement I and Statement II both are incorrect.

3) For a reaction of iodide ion with hydrogen peroxide at room temperature, what will be the order of reaction concerning iodide ion :

(1) I order

(2) Zero order

(3) II order

(4) III order

4) To find strength of given solution of $KMnO_4$ by ferrous ammonium sulphate :
(a) Oxidation of Iron
(b) Oxidation of Mn^{2+}
(c) $KMnO_4$ is used as indicator.
(d) Mohr's salt is reducing agent.

Which is correct ?

(1) a, c

(2) a, b, d

(3) b, d

(4) a, c, d

5) If the rate constant of a reaction is 0.03 s^{-1}, how much time does it take for 7.2 mol L^{-1} concentration of the reactant to get reduced to 0.9 mol L^{-1} ? (Given : log 2 = 0.301)

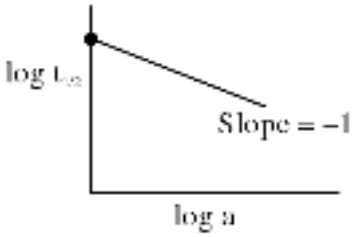
(1) 69.3 s

(2) 23.1 s

(3) 210 s

(4) 21.0 s

6) A graph between log $t_{1/2}$ and log a {a = initial concentration of reactant A} for reaction A → product is given, rate law is :



(1) $-\frac{d[A]}{dt} = K$

(2) $-\frac{d[A]}{dt} = K[A]^2$

(3) $-\frac{d[A]}{dt} = K[A]$

(4) $-\frac{d[A]}{dt} = K[A]^3$

7) For a reaction A → 2B + C the half lives are 100s and 50s when the concentration of reactant A is 0.5 and 1.0 mol L^{-1} respectively. The order of the reaction is _____.

(1) Zero

(2) One

(3) Two

(4) Three

8) **Assertion :** Rate constant determined from Arrhenius equation are fairly accurate for simple as well as complex molecules.
Reason : Reactant molecules undergo chemical change irrespective of their orientation during collision.

(1) Assertion and Reason are true but Reason is not correct explanation of Assertion.

(2) Assertion and Reason are true but Reason is correct explanation of Assertion.

(3) Both Assertion and Reason are wrong.

(4) Assertion is true and Reason is wrong

9) Correct matching of column-I with column-II is :

	Column-I		**Column-II (unit)**
(i)	G*	(a)	V
(ii)	κ	(b)	m^{-1}
(iii)	λ_{eq}	(c)	S cm^2 eq^{-1}
(iv)	E_{cell}	(d)	S cm^{-1}

(1) i-b, ii-d, iii-c, iv-a

(2) i-b, ii-c, iii-a, iv-d

(3) i-c, ii-b, iii-d, iv-a

(4) i-a, ii-c, iii-b, iv-d

10) The minimum mass of NaOH required to be added in right hand side to consume all the H^+ present in right hand side of the cell of emf, 0.70V at 25°C before its use

Zn | Zn^{+2}(0.01M) || HCl (V = 500ml) | H_2(1bar) | Pt

Given: $E^{o}_{Zn^{+2}/Zn} = -0.76V$, $\frac{2.303\,RT}{F} = 0.06$

(1) 2.0 g

(2) 4.0 g

(3) 0.2 g

(4) 0.4 g

11) If 6.43×10^5C of electricity is passed through on electrical cell containing $NaClO_3$, 245 g $NaClO_4$ are produced at the anode at the end of electrolysis. Anode efficiency is :

(1) 30%

(2) 60%

(3) 80%

(4) 95%

12) For a strong electrolyte, a plot of molar conductivity against $(\text{concentration})^{1/2}$ is a straight line, with a negative slope, the correct unit for the slope is

(1) S cm^2 $mol^{-3/2}$ $L^{1/2}$

(2) S cm^2 mol^{-1} $L^{1/2}$

(3) S cm^2 $mol^{-3/2}$ L

(4) S cm^2 $mol^{-3/2}$ $L^{-1/2}$

13) E^{o} of some elements are given as :

$I_2 + 2e^- \rightarrow 2I^{\ominus}$; $E^{o} = 0.54$ V

$MnO_4^{\ominus} + 8H^{\oplus} + 5e^- \rightarrow Mn^{+2} + 4H_2O$; $E^\circ = 1.52\,V$
$Fe^{+3} + e^- \rightarrow Fe^{+2}$; $E° = 0.77$ V
$Sn^{+4} + 2e^- \rightarrow Sn^{+2}$; $E° = 0.1$ V
Among these elements strongest reductant is :

(1) Sn^{+2}

(2) Fe^{+2}

(3) Mn^{+2}

(4) I^-

14) The molar conductivity of acetic acid at infinite dilution is 390.7 S cm^2 mol^{-1} and for 0.01 M acetic acid is 3.907 S cm^2 mol^{-1}. Calculate pH of solution :

(1) 2

(2) 3

(3) 3.5

(4) 4

15) **Assertion (A):** Molar conductivity of a weak electrolyte at infinite dilution cannot be determined experimentally.
Reason (R): Molar conductivity decreases with dilution.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is incorrect.

(4) (A) is incorrect but (R) is correct.

16) Match List-I with List-II.

	List-I		**List-II**
(A)	$Cd(s) + 2Ni(OH)_3(s) \rightarrow CdO(s) + 2Ni(OH)_2(s) + H_2O(l)$	(I)	Primary battery
(B)	$Zn(Hg) + HgO(s) \rightarrow ZnO(s) + Hg(l)$	(II)	Discharging of secondary battery
(C)	$2PbSO_4(s) + 2H_2O(l) \rightarrow Pb(s) + PbO_2(s) + 2H_2SO_4(aq)$	(III)	Fuel cell
(D)	$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$	(IV)	Charging of secondary battery

Choose the correct answer from the options given below :

(1) (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

(2) (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

(3) (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

(4) (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

17) Given :

$\Lambda_M^o[Ba(OH)_2] = 450\,S\,cm^2mol^{-1}$
$\Lambda_M^o[BaCl_2] = 240\,S\,cm^2mol^{-1}$
$\Lambda_M^o[NH_4Cl] = 130\,S\,cm^2mol^{-1}$
Calcuate Λ_M^o for NH_4OH :

(1) 225 S cm^2 mol^{-1}
(2) 235 S cm^2 mol^{-1}
(3) 190 S cm^2 mol^{-1}
(4) 200 S cm^2 mol^{-1}

18) Equivalent conductivity of $BaCl_2$, H_2SO_4 and HCl, are x_1, x_2 and x_3 S cm^2eq^{-1} at infinite dilution. If conductivity of saturated $BaSO_4$ solution is x S cm^{-1}, then K_{sp} of $BaSO_4$ is :-

(1) $\frac{500x}{(x_1 + x_2 - 2x_3)}$
(2) $\frac{10^6x^2}{(x_1 + x_2 - 2x_3)^3}$
(3) $\frac{2.5 \times 10^5x^2}{(x_1 + x_2 - x_3)^2}$
(4) $\frac{0.25x^2}{(x_1 + x_2 - x_3)^2}$

19) Consider the following statements and select the correct option :
Statement-I :- If the standard electrode potential of an electrode is negative then its reduced form is more stable compared to hydrogen gas.
Statement-II :- To measure resistance of solution, DC source is used.

(1) Both statement-I and statement-II are correct
(2) Both statement-I and Statement-II are incorrect
(3) Statement-I is incorrect and statement-II is correct
(4) Statement-I is correct and statement-II is incorrect

20) Two half cell reactions are given below
$Fe^{+2}(aq.) \rightarrow Fe^{+3}(aq.) + e^-$; $E^o_{Fe^{+2}/Fe^{+3}} = -0.77V$
$2I^-(aq.) \rightarrow I_2(s) + 2e^-$; $E^o_{I^-/I_2} = -0.54V$
then $E°_{cell}$ of cell with feasible redox reaction is :

(1) 1.31 V
(2) –0.23 V
(3) 0.23 V
(4) 1 V

21) For a cell reaction involving a two mole electron change, the standard EMF of the cell is found to be 0.295V at 25°C. The equilibrium constant of the reaction at 25°C will be.

(1) 1×10^{10}

(2) 1×10^{-10}

(3) 29.5×10^{-2}

(4) 2×10^{10}

22) How much will the potential of a hydrogen electrode change when its solution having 0.1M HCl is diluted by 100 times?

(1) Increase by 0.18 V

(2) Decrease by 0.18 V

(3) Increase by 0.12 V

(4) Decrease by 0.12 V

23) $E^{o}_{Au^{+}/Au}$ is 1.69 V and $E^{o}_{Au^{+3}/Au}$ is 1.40V than $E^{o}_{Au^{+}/Au^{+3}}$ will be :

(1) 0.19 V

(2) 2.945 V

(3) 1.255 V

(4) –1.255 V

24) The mechanism of the reaction
$2NO + O_2 \rightarrow 2NO_2$ is

$NO + NO \underset{k_2}{\overset{k_1}{\rightleftharpoons}} N_2O_2$ (fast)

$N_2O_2 + O_2 \xrightarrow{k_3} 2NO_2$ (slow)

The rate expression is :-

(1) $R = k_3 [N_2O_2] [O_2]$

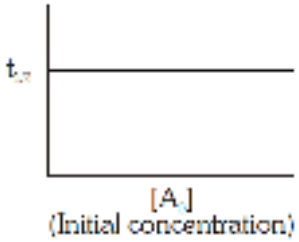
(2) $R = \frac{k_1 k_3}{k_2} [NO]^2 [O_2]^2$

(3) $R = k_1 k_3 [NO] [O_2]$

(4) $R = \frac{k_3 k_1}{k_2} [NO]^2 [O_2]^1$

25) In the following graphs, which graph(s) belong to first order

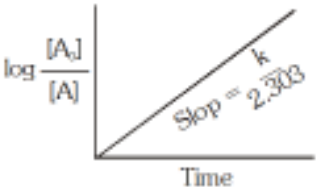
(i)



(ii)



(iii)



(iv)



(1) (i), (iii), (iv)

(2) (i), (ii), (iii)

(3) (i) only

(4) (iii), (ii), (iv)

26) What would be the rate equation of the reaction
A + B → Products
If following data is collected experimentally?

Exp.	Conc. initial [A]	Conc. initial [B]	Initial rate (r)
1	0.1 M	1.0 M	2.1×10^{-3}
2	0.2 M	1.0 M	8.4×10^{-3}
3	0.2 M	2.0 M	8.4×10^{-3}

(1) $r = k[A]^2$

(2) $r = k[B]^2$

(3) $r = k[A]^2 [B]^1$

(4) $r = k[A]^1 [B]^1$

27) **Statement-I :** Rate law for any reaction can be predicted by merely looking at the balanced chemical equation.
Statement-II : The sum of powers of the concentration of the reactants in the rate law expression is called the molecularity of that chemical reaction.

(1) Both statement I and statement II are true.

(2) Statement I is false but statement II is true.

(3) Both statement I and statement II are false.

(4) Statement I is true but statement II is false.

28) The following data were obtained during the first order gaseous reaction at constant volume : (starting with A only)
$2A(g) \rightarrow 2B(g) + C(g)$

S.No.	Time(sec.)	Total pressure (atm)
1.	0	2
2.	100	2.5

Calculate the rate constant.

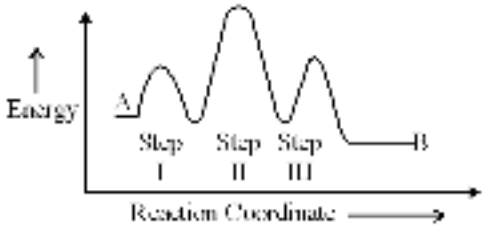
(1) $8 \times 10^{-2}\ s^{-1}$

(2) $7.5 \times 10^{-4}\ s^{-1}$

(3) $6.93 \times 10^{-3}\ s^{-1}$

(4) $6.2 \times 10^{-2}\ s^{-1}$

29) Consider the following reaction that goes from A to B in three steps as shown below:-



Choose the correct

	Number of intermediates	Number of Activated complex	Rate determining step
1	3	2	II
2	2	3	II
3	2	3	I
4	2	3	III

(1) 1

(2) 2

(3) 3

(4) 4

30) When cells of skeletal vacuoles of a frog were placed in a series of NaCl solution of differenent concentration solution at 25°C, it was observed microscopically that they remained unchanged in 0.7% solution, shrank in a more concentrated and swelled in more dilute solutions, water freezes from the 0.7% salt solution at –0.406°C. What is the osmotic pressure of the cell cytoplasm at 25°C. (K_f = 1.86 kg mol^{-1} K)

(1) 0.29 atm

(2) 2.81 atm

(3) 3.79 atm

(4) 5.34 atm

31) If 0.2 m aqueous solution of $K_4[Fe(CN)_6]$ is 100% ionised, then the freezing point of the solution will be (K_f = 1.86 K kg mol^{-1})

(1) –3.21°C

(2) –1.86°C

(3) –1.86 K

(4) –3.21 K

32) **Assertion (A) :** The osmotic pressure of 0.1 M urea solution is less than 0.1 M NaCl solution.
Reason (R) : Osmotic pressure is not a colligative property.

(1) Both Assertion and Reason are true

(2) Both Assertion and Reason are false

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

33) Which of the following statements is correct regarding a solution of two componant A and B exhibiting negative deviation from ideal behaviour?

(1) $\Delta_{mix}H \geq 0$ at constant T and P

(2) Intermolecular attractive forces between A-A and B-B are stronger than those between A-B

(3) $\Delta_{mix}V \geq 0$ at constant T and P

(4) Intermolecular attractive forces between A-A and B-B are weaker than those between A-B

34) Density of 2.05 M solution of acetic acid in water is 1.02 g/mL. The molality of the solution is :

(1) 3.28 mol kg^{-1}

(2) 2.28 mol kg^{-1}

(3) 0.44 mol kg^{-1}

(4) 1.14 mol kg^{-1}

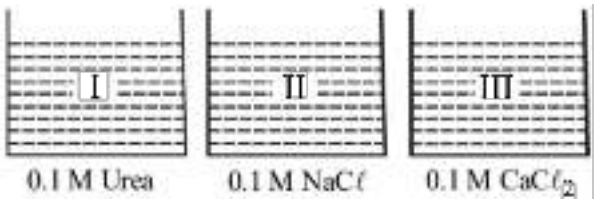
35) Compound $PdCl_4.6H_2O$ is a hydrated complex ; 1 molal aqueous solution of it has freezing point 269.28 K. Assuming 100% ionization of complex, calculate the molecular formula of the complex (K_f for water = 1.86 K kg mol^{-1}) :

(1) $[Pd(H_2O)_6]Cl_4$

(2) $[Pd(H_2O)_4Cl_2]Cl_2.2H_2O$

(3) $[Pd(H_2O)_3Cl_3]Cl.3H_2O$

(4) $[Pd(H_2O)_2Cl_4].4H_2O$

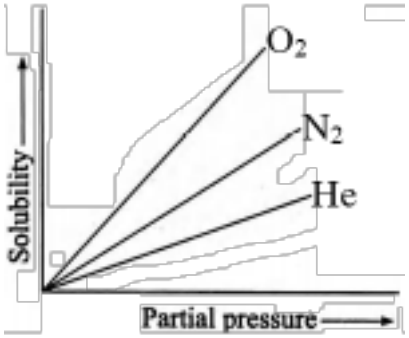
36) The correct observation is 

(1) Vapour pressure of solution I is lowest

(2) Relative lowering of vapour pressure is maximum in III

(3) Freezing point is maximum for III

(4) Boiling point is minimum for II

37) Molar solubility of helium, nitrogen and oxygen are plotted against partial pressure of the gas at constant temperature. 

Henry's law constant for these gases will lie in following sequence ?

(1) $O_2 < N_2 > He$

(2) $O_2 < N_2 < He$

(3) $O_2 = N_2 = He$

(4) $O_2 > N_2 < He$

38) At a given temperature, total vapour pressure (in Torr) of a mixture of volatile components A and B is given by

$P_{Total} = 120 - 75\ X_B$

hence, vapour pressure of pure A and B respectively (in Torr) are

(1) 120, 75

(2) 120, 195

(3) 120, 45

(4) 75, 45

39) Match list-I with list-II

	List-I		**List-II**
(A)	68% nitric acid and 32% water by mass	(I)	Minimum boiling azeotrope
(B)	Solution showing large positive deviation from Raoult's law may form	(II)	Ideal solution
(C)	n-hexane and n-heptane	(III)	Maximum boiling azeotrope
(D)	Two solutions having same osmotic pressure at a given temperature	(IV)	Isotonic solution

Choose the correct answer from the options given below :-

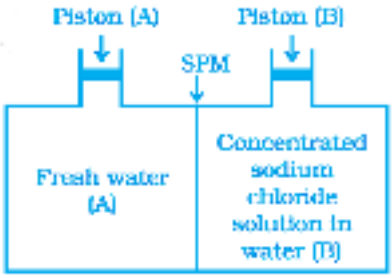
(1) (A)-I, (B)-III, (C)-II, (D)-IV

(2) (A)-III, (B)-IV, (C)-II, (D)-I

(3) (A)-III, (B)-I, (C)-IV, (D)-II

(4) (A)-III, (B)-I, (C)-II, (D)-IV

40) Consider the Fig. and mark the correct option.



(1) water will move from side (A) to side (B) if a pressure greater than osmotic pressure is applied on piston (B).

(2) water will move from side (B) to side (A) if a pressure greater than osmotic pressure is applied on piston (B).

(3) water will move from side (B) to side (A) if a pressure equal to osmotic pressure is applied on piston (B).

(4) water will move from side (A) to side (B) if pressure equal to osmotic pressure is applied on piston (A).

41) 1L of 1M HCl and 1L of 1M H_2SO_4 are completely neutralized with NaOH separately. The heat liberated in two cases are x and y calories respectively. Then which of the following relation is correct ?

(1) x = y

(2) x > y

(3) x << y

(4) y = 2x

42) Given below are two statements :
Statement I : 2F electricity is required for the oxidation of 1 mole H_2O to O_2.
Statement II : To get 40.0 g of Aluminium from molten Al_2O_3 required electricity is 4.44 F.
In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **Statement I** and **Statement II** are false.

(2) **Statement I** is true but **Statement II** is false.

(3) **Statement I** is false but **Statement II** is true.

(4) Both **Statement I** and **Statement II** are true.

43) Mass of Cu which deposits at cathode during electrolysis of $CuSO_4$ (aq.) solution, if 1.12 L O_2 (g) is evolved at anode at 1 atm and 273 K under similar conditions -
(Atomic weight of Cu = 63.5 amu)

(1) 1.12 g

(2) 2.24 g

(3) 6.35 g

(4) 63.5 g

44) Three moles of electrons are passed through three aqueous solutions in succession containing $AgNO_3$, $CuSO_4$ and $AlCl_3$, respectively. The molar ratio of amounts of cations reduced at cathode will be:-

(1) 6 : 3 : 2
(2) 2 : 1 : 3
(3) 3 : 2 : 1
(4) 6 : 3 : 0

45) For Arrhenius equation when a graph is plotted between log K Vs $\frac{1}{T}$ slope obtained is –0.5. The activation energy is given by :-

(1) 5.9 cal
(2) 7.9 cal
(3) 1.2 cal
(4) 2.3 cal

BIOLOGY

1) Which of the following statement is incorrect about apomixis ?

(1) It is a type of sexual reproduction, which mimics asexual reproduction.
(2) It is a type of asexual reproduction but mimics sexual reproduction.
(3) It is a type of reproduction which produce seed without fertilization.
(4) This type of reproduction does not leads to variation in progeny.

2) Select the incorrect statement -

(1) Pollination by wind is more common amongst abiotic pollinations.
(2) Embryo develops at micropylar end of the embryo sac where the zygote is situated.
(3) In dicot embryo, the root tip is covered with coleorhiza.
(4) The embryo may enter a state of inactivity called dormancy.

3) Select the incorrect statement -

(1) The zygote gives rise to the proembryo and subsequentially to the globular heart shaped and mature embryo
(2) Endosperm in castor is completely consumed during embryo development.
(3) Embryo of monocotyledons posses only one cotyledon.
(4) In dicot embryo, the root tip is covered with root cap.

4) **Assertion (A) :** Each cell of sporogenous tissue in microsporangium is potential pollen or microspore mother cell.

Reason (R) : Each cell of the sporogenous tissue is capable of giving rise to a microspore tetrad.

(1) (A) is true but (R) is false.

(2) Both (A) & (R) true but (R) is not correct explanation of (A)

(3) Both (A) and (R) are incorrect.

(4) Both (A) & (R) true and (R) is correct explanation of (A).

5)

Given below are two statements, one is labelled as (A) Assertion an other is Reason (R).
Assertion : Endosperm development precedes embryo development.
Reason : The primary endosperm cell divides repeatedly and forms a triploid endosperm tissue.
Select the correct option for above statements.

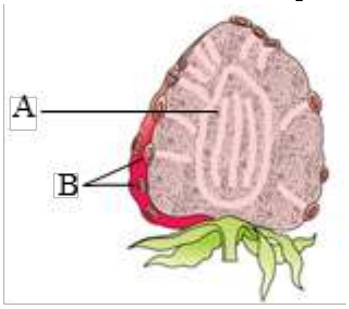
(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

6) Select the correct option in which labelled parts (A, B) correctly identified ?



(1) A-Thalamus, B-Achene

(2) A-Mesocarp, B-Endocarp

(3) A-Achene, B-Mesocarp

(4) A-Cotyledon, B-Seed coat

7) In angiosperms the correct sequence of events in formation of female gametophyte in the ovule is :
(A) Three sequential mitotic divisions.
(B) Degeneration of 3 megaspores.
(C) Production of four megaspore through meiosis.
(D) Three cell grouped together to form egg apparatus.
(E) 8-nucleate, 7-celled matured embryo sac.
Choose the correct answer :

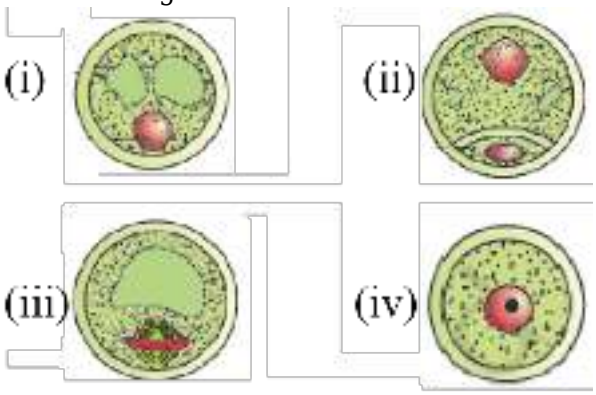
(1) (A), (B), (C), (D), (E)

(2) (C), (E), (A), (D), (B)

(3) (B), (C), (A), (D), (E)

(4) (C), (B), (A), (D), (E)

8) Few events of male gametophyte development in angiosperm are shown in figure given below. Arrange the diagram in ascending order of their occurrence and select the correct order from



options given below

(1) i, ii, iii, iv

(2) i, iv, iii, ii

(3) i, ii, iv, iii

(4) iv, i, iii, ii

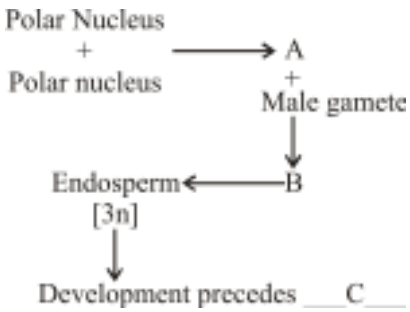
9) How many pollen grains are produced by single flower in *Solanum nigrum* ? (If each microsporangium have 98 microspore mother cells) :-

(1) 5840

(2) 7840

(3) 9840

(4) 58400



10) What are A, B and C are

	A	B	C
(1)	Definitive nucleus	X-body	Embryogeny
(2)	Zygote	Synergid cell	Megasporogenesis
(3)	Synergid	Primary endosperm Nucleus	Syngamy
(4)	Secondary nucleus	Primary endosperm nucleus	Embryogeny

(1) 1

(2) 2

(3) 3

(4) 4

11) Read the following statement and find out which one is not correct ?

(1) When a flower has both androecium and gynoecium, it is called bisexual.

(2) In mango and coconut, the fruit is known as a drupe.

(3) Sterile stamen is called staminode

(4) Dianthus and primrose are not example of free central placentation.

12) Some characters are given below, which of the following is not correct about seeds of maize?
(A) Seed coat is membranous and generally fused with fruit wall.
(B) The outer covering of endosperm separates embryo by aleurone layer.
(C) The plumule and radicle are not covered with coleoptile and coleorhiza respectively.
(D) Embryo consists of two large shield shaped cotyledons known as scutellum.

(1) A and B

(2) only C

(3) only D

(4) C and D

13) Which one of the following options gives the **correct** categorisation of six plants according to the type of placentation (a,b,c) they give out ?

	(a) Axile	**(b) Basal**	**(c) Parietal**
(1)	China rose, lemon	Mustard, wheat	Sun flower, *Argemone*
(2)	Sun flower, lemon	China rose, wheat	Mustard, *Argemone*
(3)	China rose, *Argemone*	Sun flower, wheat	Mustard, lemon
(4)	China rose, lemon	Sun flower, marigold	Mustard, *Argemone*

(1) 1

(2) 2

(3) 3

(4) 4

14) How many plants in the given below have petals with twisted aestivation ?
Calotropis, Mustard, China rose, Soyabean, Cotton, Petunia, Tulip, Lady finger, Cassia, Gulmohar

(1) Six

(2) Four

(3) Five

(4) Three

15) Match the following and choose correct option.

	Group A		Group B
(A)	Monoadelphous androecium	(i)	Lily
(B)	Diadelphous androecium	(ii)	Brinjal
(C)	Epipetalous stamens	(iii)	Pea
(D)	Epiphyllous stamens	(iv)	China rose

Option :-

	A	**B**	**C**	**D**
(1)	(iv)	(iii)	(i)	(ii)
(2)	(iv)	(iii)	(ii)	(i)
(3)	(iii)	(iv)	(ii)	(i)
(4)	(iii)	(ii)	(iii)	(i)

(1) 1

(2) 2

(3) 3

(4) 4

16) Which one of the plant groups contain modified adventitious root ?

(1) Mustard, Banyan, Wheat

(2) *Monstera*, Banyan, Grass

(3) Carrot, Grass, Mustard

(4) Mustard, Grass, Banyan

17) The stem develops from

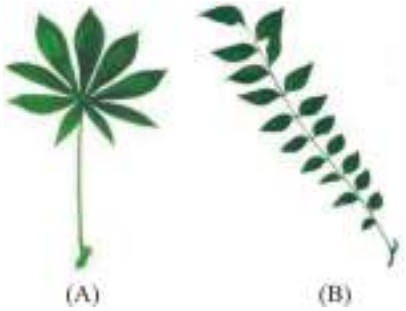
(1) hypocotyl

(2) epicotyl

(3) radicle

(4) plumule

18) The two sketches A and B given below represents two different types of leaves. Select the option



in which both are correctly identified :-

	A	**B**
(1)	Simple leaf	Compound leaf
(2)	Compound leaf	Simple leaf
(3)	Pinnately compound leaf	Palmately compound leaf
(4)	Palmately compound leaf	Pinnately compound leaf

(1) 1

(2) 2

(3) 3

(4) 4

19) Sheathing leaf base is found in :-

(1) Legumes

(2) Monocots

(3) Dicots

(4) All Angiosperms

20) Which of the following character is not related to racemose inflorescence?

(1) The main axis continues to grow

(2) Flowers are borne laterally

(3) Acropetal succession present

(4) Main axis terminates in a flower



21) The following diagram represents the position of ovary in which plant:-

(1) Brinjal

(2) Guava

(3) Rose

(4) Mustard

22) **Statement- I :** Mustard is an example of hypogynous flower.

Statement- II : Mustard is an example of inferior ovary.

(1) Both Statements I and II are correct
(2) Both Statements I and II are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

23) Ovules are attached on central axis in unilocular ovary in :-

(1) Axile placentation
(2) Parietal placentation
(3) Free central placentation
(4) Marginal placentation

24) Match the column-I to column-II, III and find out correct option.

Column-I		**Column-II**		**Column-III**	
I	Hypogynous	A	Ovary half inferior	S	Chinarose, brinjal
II	Perigynous	B	Overy superior	T	Guava, cucumber
III	Epigynous	C	Overy inferior	U	Plum, peach

(1) I-B-S, II-A-U, III-C-T
(2) I-A-S, II-B-T, III-C-U
(3) I-C-U, II-A-S, III-B-T
(4) I-A-S, II-C-U, III-B-T

25) Select correct combination of column I, II & III.

	Column I	**Column II**	**Column III**
(A)	Marginal	 (I)	(1) Sunflower, Marigold
(B)	Axile	 (II)	(2) *Dianthus*, Primrose
(C)	Parietal	 (III)	(3) Mustard, *Argemone*

(D)	Free Central	(IV) 	(4) China rose, Tomato, Lemon
(E)	Basal	(V) 	(5) Pea

(1) A-V-(5); B-II-(4); C-I-(3); D-III-(2); E-IV-(1)
(2) A-I-(5); B-II-(4); C-III-(3); D-IV-(2); E-IV-(1)
(3) A-V-(1); B-II-(4); C-I-(2); D-III-(3); E-IV-(5)
(4) A-V-(1); B-III-(2); C-II-(4); D-I-(5); E-IV-(3)

26) Which of the following is not related to the structure of stomata ?

(1) Stomatal aperture
(2) Guard cells
(3) Bulliform cells
(4) Subsidiary cells

27)

How many in the list given below are the part of ground tissue system :

Epidermis, Root hairs, Xylem, Phloem, Cortex, Pericycle, Pith
(1) One
(2) Two
(3) Three
(4) Four

28) Conjoint vascular bundles are common in

(1) Root
(2) Stem
(3) Leaf
(4) Both (2) & (3)

29) The cross section of a plant material shows the following anatomical features under microscope:-

The vascular bundles are radially arranged, eight xylem strands with exarch condition and well developed pith.
The plant material should be :-

(1) Dicot stem

(2) Monocot stem

(3) Dicot leaf

(4) Monocot root

30) Which of the following is false about monocot stem ?
(I) Vascular bundles are conjoint, collateral and surrounded by parenchymatous sheath
(II) Hypodermis is sclerenchymatous
(III) Ground tissue is differentiated in cortex, endodermis and pith

(1) Only I

(2) I and II

(3) I and III

(4) Only II

31) Which of the following is responsible for leaf rolling and unrolling in monocot :

(1) Bundle sheath

(2) Bulliform cell

(3) Spongy tissue

(4) Palisade tissue

32) **Statement-I :** The adaxial epidermis of dorsiventral leaf has more stomata than abaxial epidermis.
Statement-II : In grasses, certain abaxial epidermal cells along the veins are modified into large, empty and colourless cells known as bulliform cells.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II are incorrect.

33) **Statement-I :** Mesophyll, which possesses chloroplasts and carry out photosynthesis, is made up of collenchyma.
Statement-II : The tissue between the upper and the lower epidermis is called the mesophyll.

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement-II is correct

34) **Assertion :** In isobilateral leaf stomata are equally present on both the surfaces.
Reason : Mesophyll is not differentiated into palisade and spongy parenchyma in monocot leaf.

(1) Both assertion and reason are false.

(2) Assertion is true, reason is false.

(3) Both assertion and reason are true but reason is not correct explanation of assertion.
(4) Both assertion and reason are true and reason is correct explanation of assertion.

35) Which of the following is incorrect about isobilateral leaf ?

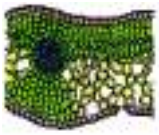
(1) Vascular bundles are conjoint and closed.
(2) Xylem faces adaxial surface while phloem faces abaxial surface.
(3) Parallel venation found in monocot leaves.
(4) Varible sized vascular bundles are present all over leaf.

36)

Which of the following tissue is not a part of stele ?

(1) Pericycle
(2) Vascular bundle
(3) Pith
(4) Endodermis

37) Match the column-I with column-II and select the correct option given below :-

	Column-I		**Column-II**
(A)		(P)	Dicot root
(B)		(Q)	Dicot leaf
(C)		(R)	Monocot leaf
(D)		(S)	Maize stem
		(T)	Monocot root

(1) A : Q; B : S; C : T; D : R
(2) A : Q; B : S; C : P; D : R
(3) A : R; B : S; C : T; D : Q
(4) A : R; B : S; C : P; D : Q

38) Which of the following is not part of the inner most whorl of the flower ?

(1) Stamen

(2) Style

(3) Ovary

(4) Stigma

39) Read the statements carefully :-
(A) Flowers are morphological and embryological marvel and the site of asexual reproduction.
(B) Flowering is initiated by several hormonal and structural changes.
(C) The number and length of stamens are variable in flowers of different species.
(D) Four lobes are present in one anther
How many above statements are correct :-

(1) One

(2) Three

(3) Two

(4) Four

40) In a typical stamen the proximal end of the filament is :-

(1) Attached to the thalamus

(2) Generally bilobed structure

(3) Towards the stigma

(4) Pollen grains producing part

41) Which of the following is wrongly matched ?

(1) Pollensacs – Microsporangia

(2) Stamen – Microsporophyll

(3) Pollengrains – Male gametophyte

(4) Ovule – Megasporophyll

42) Viability of pollen grain in rice is

(1) With in 60 min

(2) With in 30 min

(3) With in 30 seconds

(4) With in 30 days

43)

Where the development of dicot embryo occurs at :-

(1) Chalazal end of the embryo sac

(2) Micropylar end of the embryo sac

(3) Centre of embryo sac

(4) Antipodal region of embryo sac

44) Ploidy of endosperm will be ..... if the male & female parents are hexaploid and tetrapoid respectively :

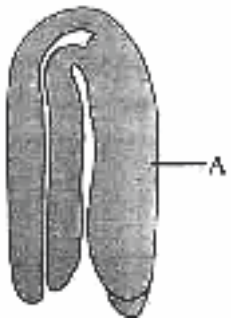
(1) 8x

(2) 7x

(3) 16x

(4) 10x

45) Identify 'A' in the given diagram :



(1) Cotyledons

(2) Radicle

(3) Hypocotyl

(4) Epicotyl

46) Choose the correct path of milk :-

(1) Mammary alveolus → Mammary tubule → Mammary duct → Mammary ampulla → Lactiferous duct

(2) Mammary alveolus → Mammary duct → Mammary tubule → Mammary ampulla → Lactiferous duct

(3) Mammary alveolus → Mammary tubule → Mammary ampulla → Mammary duct → Lactiferous duct

(4) Mammary alveolus → Mammary tubule → Mammary ampulla → Lactiferous duct → Mammary duct

47) Choose the incorrect :-

(1) Spermatogonia $\xrightarrow[\text{Growth}]{\text{Mitosis}}$ primary spermatocytes

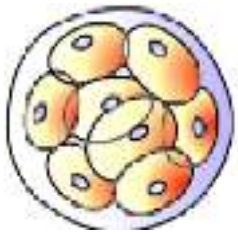
(2) Secondary spermatocytes $\xrightarrow[\text{division}]{2^{nd}\text{ meiotic}}$ spermatids

(3) Primary spermatocytes $\xrightarrow[\text{division}]{\text{Reductional}}$ secondary spermatocytes

(4) Spermatids $\xrightarrow[\text{division}]{\text{Reductional}}$ spermatozoa

48) Among the following identify the structure which is involved in implantation:-

(1) 

(2) 

(3) 

(4) 

49) Given below are some statements :-
(A) Oogonia multiplies by mitotic division and increases in numbers after puberty.
(B) Tertiary follicle is characterised by a fluid filled cavity called antrum.
(C) Meiotic division during oogenesis is different from that in spermatogenesis.
(D) Spermatogonia have 46 chromosomes and always undergo meiotic cell division.
(E) Spermatozoa are transformed into spermatids by the process called spermiogenesis
Select the set of correct statements :-

(1) C and D only

(2) A, D and E only

(3) B and C only

(4) B and E only

50) Match column-I with column-II.

	Column-I		**Column-II**
A.	Labia minora	i.	Cushion of fatty tissue
B.	Mons pubis	ii.	Upper junction of labia minora
C.	Clitoris	iii.	Paired fold of tissue
D.	Hymen	iv.	Membrane of mucosal tissue

(1) A-i, B-ii, C-iii, D-iv
(2) A-iii, B-i, C-ii, D-iv
(3) A-ii, B-iii, C-i, D-iv
(4) A-iii, B-iv, C-i, D-ii

51) Match the column-I with column-II.

	Column-I		**Column-II**
(A)	LH	I.	Oxytocin
(B)	Corpus luteum	II.	Ovulation
(C)	Posterior pituitary	III.	Relaxin
(D)	Placenta	IV.	Progesterone

(1) A-IV, B-II, C-III, D-I
(2) A-IV, B-II, C-I, D-III
(3) A-II, B-IV, C-III, D-I
(4) A-II, B-IV, C-I, D-III

52) Which of the following structure has erectile tissue ?

(1) Labia minora
(2) Labia majora
(3) Clitoris
(4) Hymen

53) Muscles associated with mammary glands are-

(1) Biceps
(2) Pectoralis majors
(3) Gluteus maximum
(4) Triceps

54) Some factors which help in the process of spermiogenesis are secreted by ?

(1) Leydig cell
(2) Sertoli cell
(3) Interstitial cell
(4) Gonads

55) Which of the following hormone is responsible for secretion of androgens?

(1) F.S.H
(2) L.H
(3) Estrogen
(4) A.B.P

56) Which placental hormone act as LH and maintain the corpus luteum in ovary of a pregnant female?

(1) Relaxin
(2) hCG
(3) hPL
(4) Progesterone

57) What would be the number of chromosome in spermatids?

(1) 46
(2) 23
(3) 26
(4) 22

58) Which part of sperm is source of energy for swimming?

(1) Head
(2) Neck
(3) Middle piece
(4) Tail

59) The primary oocyte completes its first meiotic division within :-

(1) Graafian follicle
(2) Secondary follicle
(3) Fallopian tube
(4) Tertiary follicle

60) Which hormone is not produced in women only during pregnancy ?

(1) Relaxin
(2) HCG
(3) HPL
(4) Progesterone

61) Meiosis-I of primary oocyte arrested in which stage ?

(1) Metaphase-II
(2) Prophase-II
(3) Prophase-I
(4) Metaphase-I

62) The first movements of foetus are usually observed during the ?

(1) One month

(2) 12 Weeks

(3) Fifth month

(4) Second month

63) Limbs and external genital organs are well developed, by the end of ?

(1) 12 weeks

(2) 2 weeks

(3) 24 weeks

(4) Fifth month

64) **Statement-I :-** Humans are sexually reproducing & viviparous.
Statement-II :- There are no remarkable difference between the reproductive events in male & female.

(1) Statement-I and II both are correct.

(2) Statement-I and II both are incorrect.

(3) Only Statement-I is correct.

(4) Only Statement-II is correct.

65) Select incorrect option :-

(1) Ovum and sperm transported to the ampullary region during fertilisatin.

(2) During fertilisation meiotic-II is completed and 2nd bar body is released.

(3) There are 46 chromosomes in human zygote.

(4) Both (1) & (2)

66) **Assertion (A) :-** The pregnancy continuous even after the ovaries are removed once the placenta is fully developed.
Reason (R) :- After development of placenta estrogen and progesterone are secreted mainly by the placenta.

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true but R is NOT correct explanation of A.

(4) Both A and R are true and R is correct explanation of A.

67) Match the column-I with the column-II

	Column-I		**Column-II**
A.	Ovary	i.	Through which milk is sucked out from mammary gland
B.	Seminiferous tubules	ii.	In which sperms are produced
C.	Uterus	iii.	Produce several steroid hormones
D.	Lactiferous duct	iv.	Site of follicular development

		v.	Inner layer shows cyclical changes during menstrual cycle
		vi.	Cells of it influenced by FSH

Options :-

(1) A-iv, ii; B-ii, iii; C-i; D-v

(2) A-iii, iv; B-ii, vi; C-v; D-i

(3) A-iii, iv; B-i; C-v,vi; D-ii

(4) A-iii,iv,vi; B-i; C-v; D-ii

68) **Assertion (A) :** The placenta facilitate the supply of oxygen and nutrient to the embryo and also removal of carbon dioxide and excretory/waste materials produced by the embryo.
Reason (R) : The placenta is connected to the embryo through an umbilical cord which help in the transport of substances to and from the embryo.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is not correct.

(4) (A) is incorrect but (R) is correct.

69) **Statement-I :** The secretion of gonadotropins (LH and FSH) increase gradually during the follicular phase and stimulates follicular development as well as secretion of high level of progestrone by the growing follicles.
Statement-II : The corpus luteum secrete large amount of progestrone which is essential for maintenance of the endocardium of uterus.

(1) Statement-I and statement-II both are correct.

(2) Statement-I is correct, but statement-II is incorrect.

(3) Statement-I is incorrect, but statement-II is correct.

(4) Both statement-I and II are incorrect.

70) The female external genitalia include :-
(a) Mons pubis
(b) Labia majora
(c) Hymen
(d) Clitoris
(e) Labia minora
(f) Ovary
(g) Uterus

(1) a, b, c, d, e

(2) a, c, d, e, f, g

(3) c, d, e, f, g

(4) a, b, f, g

71) Select the correct sequence of transport of sperm cells in male reproductive system :-

(1) Testis → Epididymis → Vasa efferentia → Rete testis → Inguinal canal → Urethra

(2) Seminiferous tubules → Rete testis → Vasa efferentia → Epididymis → Vas deferens → Ejaculatory duct → Urethra → Urethral meatus

(3) Seminiferous tubules → Vasa deferens → Epididymis → Inguinal canal → Urethra

(4) Testis → Epididymis → Vasa efferentia → Vas deferens → Ejaculatory duct → Inguinal canal → Urethra → Urethral meatus

72) Which of the following statement is correct about testis ?

(1) Testis is circular in shape

(2) Each testis have 300 compartment called testicular lobules.

(3) Length of testis is about 4 to 5 cm width of testis 2 to 3 cm.

(4) Each lobule contain 3 to 5 highly coiled seminiferous tubules.

73) Each ovary is connected to the pelvic wall and the uterus by :-

(1) Tendons

(2) Ligaments

(3) Adipose tissue

(4) Areolar tissue

74) Which of the following contraceptive methods can prevent ovulation?

(1) CuT, Condoms

(2) Combination of estrogen and progesterone taken orally

(3) Diaphragms, cervical cap and vaults

(4) Coitus interruptus and lactational amenorrhoea

75) As per the medical termination of pregnancy (Amendment) Act, 2017 - if the pregnancy has lasted more than 12 weeks, but fewer than 24 weeks, _X_ registered medical practitioners must be of the opinion, formed in good faith, that the required ground exist here X is :-

(1) One

(2) Three

(3) Two

(4) Four

76) If a person suffers from very low sperm counts, which ART could be used?

(1) ZIFT

(2) AI

(3) IUT

(4) GIFT

77) Which of the following function is performed by multiload 375 ?

(1) Inhibit ovulation

(2) Make the uterus unsuitable for implantation

(3) Decrease activity of phagocytic cells of uterus

(4) Suppress sperm motility and fertilising capacity of sperm

78) Identify the incorrect match:

(1) Saheli – Non steroidal pill

(2) Female condom – Barrier method of contraception

(3) Multiload 375 – Non-medicated IUD

(4) Progestasert – Hormone releasing IUD

79) Given below are four statements (a-d) regarding assisted reproductive technologies:-
(a) **ZIFT** - The zygote or early embryo (with upto 8 blastomere) transferred into the fallopian tube
(b) **ICSI** - A sperm is directly injected into the fallopian tube to form an embryo in the laboratory
(c) **IUI** - The semen collected either from the husband or healthy donor is artificially introduced into the uterus
(d) **GIFT** - Transfer of zygote collected from a donor into the fallopian tube of another female who cannot produce one but can provide suitable environment for fertilisation and development
Which two of the above statements are correct?

(1) a & d

(2) b & c

(3) a & c

(4) b & d

80) **Assertion :** Amniocentesis is usually done when a woman is between 14-16 weeks pregnant.
Reason : Statutory Ban on amniocentesis checks increasing menace of female foeticide.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

81) Identify the contraceptive related to hormones:-

(1) 

(2) 

(3) 

(4) 

82) If male is impotent and female is fertile then what possible assisted reproductive technology is used for treating the infertility in particular couple?

(1) GIFT

(2) Intrauterine insemination

(3) ZIFT

(4) Embryo transfer

83) In lactational amenorrhea, ovulation does not occur during the period of intense lactation because :-

(1) Presence of large amount of IgA in milk

(2) Increase in estrogen and progestrone ratio

(3) High level of prolactin

(4) Synthesis of prostaglandin

84) Which of the following contraceptive devices makes the uterus unsuitable for implantation & the cervix hostile to the sperms ?

(1) Progestasert

(2) Vault

(3) Condoms

(4) Cervical caps

85) In the given contraceptive methods, how many are Natural methods ? Periodic abstinence, coitus interrupts, barrier method, lactational amenorrhea, Diaphragms, vaults

(1) 6

(2) 5

(3) 4

(4) 3

86) If female undergoing tubectomy, which one of the following event will not occur ?

(1) Menstruation cycle

(2) Ovulation

(3) Fusion of sperm and ovum

(4) Formation of graffian follicle

87) Regarding test tube baby programme, select the correct sequential steps.
(A) Completion of further development of zygote/embryo
(B) Formation of zygote under simulated conditions in the laboratory
(C) Collection of ova from the wife/donor (female) and sperms from the husband/donor (male)
(D) Transfer of zygote/embryo into the fallopian tube/uterus
Choose the correct answer from the options given below.

(1) A, D, C, B

(2) C, B, D, A

(3) B, A, D, C

(4) D, A, C, B

88)

Match the following :

(A)	Non-medicated IUDs	(a)	Cu-T
(B)	Copper releasing IUDs	(b)	Saheli
(C)	Hormone releasing IUDs	(c)	Lippes loop
(D)	Non-steroidal pill	(d)	LNG-20

(1) A – b, B – c, C – d, D – a

(2) A – c, B – a, C – b, D – d

(3) A – c, B – a, C – d, D – b

(4) A – a, B – c, C – d, D – b

89) Implants contain :

(1) Progestogens alone only

(2) Estrogens alone only

(3) Progestogens alone or in combination with estrogen

(4) Multi load-375

90)

Which of the following contraceptive methods also helps in preventing STDs ?

(1) Copper-T

(2) Oral pills

(3) Condom

(4) IUD

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	4	2	1	1	1	2	2	1	1	2	1	3	2	4	4	3	3	4	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	3	3	1	4	4	3	2	3	3	1	2	2	3	4	1	4	1	2	1
Q.	41	42	43	44	45															
A.	1	3	3	4	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	3	1	4	1	2	3	3	1	3	2	1	1	4	3	3	2	3	2	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	4	4	4	1	1	3	3	2	4	2	3	4	2	3	2	2	3	4	2
Q.	86	87	88	89	90															
A.	4	4	3	4	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	3	2	4	2	1	4	4	2	4	4	4	4	4	2	2	4	4	2	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	3	3	1	1	3	3	4	4	3	2	4	4	3	4	4	1	1	3	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	2	2	3	1	4	3	3	2	4	3	2	2	2	2	2	3	4	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	3	1	3	2	4	2	1	4	1	2	3	2	2	3	2	4	3	3	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	2	3	1	4	3	2	3	3	3										

SOLUTIONS

PHYSICS

1)

To measure current (A) should be in series and to measure potential (V) should be in parallel.

2) $E = \rho J \Rightarrow E_A < E_B < E_C$
so magnitude of potential gradient
$E = -\frac{dV}{dr}$ = magnitude of slope m
Magnitude of slope
$m_A < m_B < m_C$

3)

$H = \frac{V^2}{R}$; reach $= \frac{R}{n} = R_{net} = \frac{R}{n^2}$

$H^1 = \frac{V^2}{Req} = \frac{V^2}{\frac{R}{n^2}} = n^2\left(\frac{V^2}{R}\right) = n^2H$

4) $I = ncAv_d$

$v_d \propto \frac{1}{A} \propto \frac{1}{r^2}$

$\frac{v_{d_{left}}}{v_{d_{right}}} = \frac{r^2_{right}}{r^2_{left}} = \left(\frac{1}{3}\right)^2$

$v_{d\,left} = \frac{v_{d_{right}}}{9} = \frac{0.50 \times 10^{-4}}{9} m/s = 0.055 \times 10^{-4}$ m/s

5) $R = \frac{\rho\ell}{A} = \left(\frac{m}{ne^2\tau}\right)\frac{\ell}{A}$

6) $i = \frac{\varepsilon}{R + r}$

$2 = \frac{\varepsilon}{r + 3}$ ...(1)

$1.25 = \frac{\varepsilon}{r + 6}$ ...(2)

On solving $r = 2\Omega$, $\varepsilon = 10$ V.

7)

Use KCL

8) $R = \frac{\rho \ell}{A}$

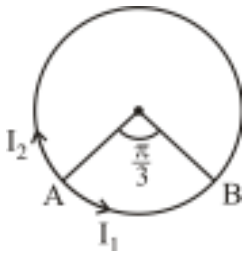
$D = \frac{M}{V}$

$A = \frac{M}{\ell D}$

$R \alpha \frac{\ell^2}{M}$

$V_1 : V_2 : V_3 = \frac{\ell_1^2}{M_1} : \frac{\ell_2^2}{M_2} : \frac{\ell_3^2}{M_3}$

$V_1 : V_2 : V_3 = 4 : 3 : 8$



9)

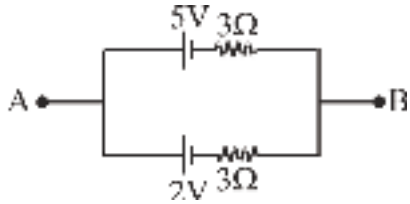
2π —— R

1 —— $\frac{R}{2\pi}$

So for $\frac{\pi}{3}$ radian = $\frac{R}{2\pi} \times \frac{\pi}{3} = \frac{R}{6}\Omega$

Similarly for $\frac{5\pi}{3}$ radian = $\frac{5R}{6}$

$R_{AB} = \frac{\frac{R}{6} \times \frac{5R}{6}}{\frac{R}{6} + \frac{5R}{6}} = \frac{5R}{36}$



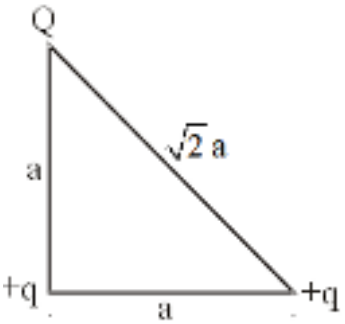
10)

$E_{eq} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}}$

$\Rightarrow E_{eq} = \frac{\frac{5}{3} + \frac{2}{3}}{\frac{1}{3} + \frac{1}{3}} = 3.5V$

or $\frac{V-5}{3} + \frac{V-2}{3} = 0$ V = 3.5 V

11) $U = k\left[\frac{q^2}{a} + \frac{Qq}{a} + \frac{Qq}{\sqrt{2}a}\right] = 0$



$$q = -Q\left[1 + \frac{1}{\sqrt{2}}\right]$$

$$Q = \frac{-q\sqrt{2}}{\sqrt{2}+1} \quad \Rightarrow Q = \frac{-2q}{2+\sqrt{2}}$$

12) The sign of flux is positive through right end cap.

13)

1. $E_{in} = \frac{KQr}{R^3}$
2. $E_{out} = \frac{KQ}{r^2}$

Solution :

At r = 20 cm $\Rightarrow E = \frac{kQ}{r^2} = 100 V/m \Rightarrow kQ = 100r^2$ ....(i)

At r' = 3 cm $\Rightarrow E = \frac{kQr}{R^3} = \frac{(100r^2)r'}{R^3}$

$$= \frac{100 \times (20 \times 10^{-2})^2 \times 3 \times 10^{-2}}{(10 \times 10^{-2})^3} = 120 V/m$$

14)

Here, $P_s = \frac{4T}{r}$

and $P_e = \frac{\sigma^2}{2\varepsilon_0}$

Since bubble is in equilibrium

$P_e = P_s$

$$\frac{\sigma^2}{2\varepsilon_0} = \frac{4T}{r} \Rightarrow \sigma = \sqrt{\frac{8T\varepsilon_0}{r}}$$

The potential V of a spherical conductor is given by $\frac{\sigma r}{\varepsilon_0}$

$$V = \frac{r}{\varepsilon_0} \times \sqrt{\frac{8T\varepsilon_0}{r}}$$

$$V = \sqrt{\frac{r^2}{\varepsilon_0^2} \times \frac{8T\varepsilon_0}{r}}$$

$$V = \sqrt{\frac{8Tr}{\varepsilon_0}}$$

15) Electric potential difference between two shells:

$$V_1 - V_2 = kQ_1\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

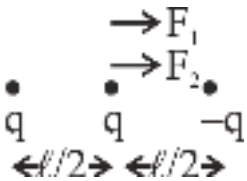
Where Q_1 is the charge on the inner shell, and R_1, R_2 are the radii of the inner and outer shells. The potential difference between the shells depends only on the charge of the inner shell (Q_1) and the radii (R_1 and R_2). Since the electric field between the shells depends solely on Q_1, the potential difference does not change when the charge on the outer shell is varied.
Both Assertion (A) and Reason (R) are correct, and the Reason correctly explains the Assertion.

16) A + B + C = X
A + B + 2 = 10
A + B = 8C

17)



$$F = \frac{kq^2}{\ell^2}$$



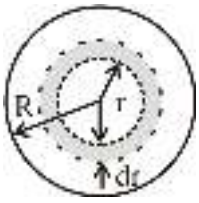
$F_{net} = F_1 + F_2$ towards –θ

$$= \frac{kq^2}{(\ell/2)^2} + \frac{kq^2}{(\ell/2)^2}$$

$$= 8\frac{kq^2}{\ell^2} = 8F$$

18) In electrostatics condition only, any system of charge particles cannot remain in stable equilibrium.

19) Let us assume an elemental ring of radius 'r' & thickness 'dr', then charge on ring



dQ = σ (Area of shaded region)
= σ (2πrdr)

$$\int dQ = \int \sigma_0 \frac{r^2}{R^2} 2\pi r\, dr \Rightarrow Q = \frac{\pi \sigma_0 R^2}{2}$$

Hence option (4)

20)

Radius of shell R = 2cm

Total charge on shell $Q = \sigma \times 4\pi R^2$

$= 10^{-4} \times 4\pi \times (2)^2$

$= 16\pi \times 10^{-4} C$

The point is at a distance r = R + 2 = 4 cm from the center

Potential at distance r is $V = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r}$

$$V = \left(9 \times 10^9\right) \times \frac{16\pi \times 10^{-4}}{4 \times 10^{-2}} = 36\pi \times 10^7 V$$

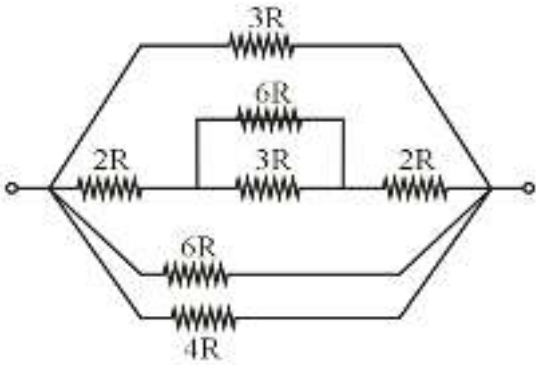
Work done W = q × V

$= (40 \times 10^{-9}) \times (36\pi \times 10^7)$

$= 1440\pi \times 10^{-2}$ J

W = 14.4 π Joule

21) Opening the upper node and removing the two resistors from the middle branches, we get.



Solving we get $R_{eq} = \frac{12}{11}R$.

22)

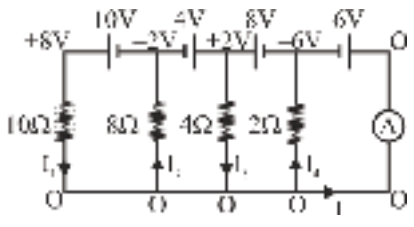
As $\rho = \rho_0(1 + \propto \Delta T)$

$$\propto = \frac{\Delta\rho}{\rho_0 \Delta T} \Rightarrow 4 \times 10^{-4} = \frac{2 \times 10^{-8}}{\rho_0(50)}$$

$$\Rightarrow \rho^0 = \frac{2 \times 10^{-8}}{4 \times 10^{-4} \times 50}$$

$\rho_0 = 10^{-6}\ \Omega$ – m

23)



$I = I_1 - I_2 + I_3 - I_4$

$= \frac{8}{10} - \frac{2}{8} + \frac{2}{4} - \frac{6}{2}$ I = –1.95 A, magnitude of current in ammeter is = 1.95 A

24) $R_S = \frac{V}{I_g} - G$

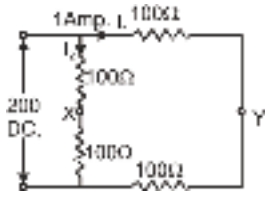
for 1V range ⇒ $R_s = \frac{1}{10^{-3}} - 50 = 950\ \Omega$

⇒ $R_1 = 950\ \Omega$

for 10 V range $R_s = \frac{10}{10^{-3}} - 50 = 10 \times 10^3 - 50$

$= 9950\ \Omega$

∴ $R_2 = 9950 - 950 = 9000\ \Omega = 9k\Omega$

25) 

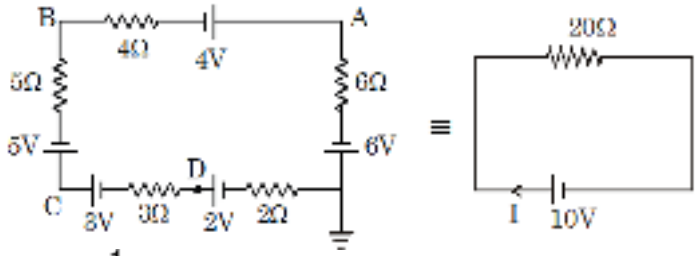
$R_{eq} = \frac{200 \times 200}{200 + 200} = 100\Omega$

Net current I = $\frac{200}{100}$ = 2 Amp.

$I_2 = I_1 = = 1$ Amp.

So $V_x - V_y = 100 - 100 = 0$

26)



⇒ $I = \frac{1}{2}A$

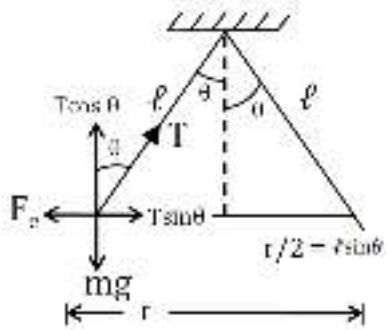
$V_A - 6 \times 1/2 + 6 = 0 \Rightarrow V_A = -3V$

$V_B - 4 \times 1/2 + 4 = V_A = -3 \Rightarrow V_B = -5V$

$0 - 2 \times 1/2 + 2 - 3 \times 1/2 + 3 = V_C = 2.5V$

$0 - 2 \times 1/2 + 2 = V_D \Rightarrow V_D = 1V$

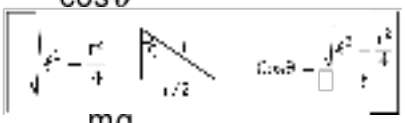
27) Bulbs are identical & connect in parallel so V same but i become twice



28)

$$T\cos\theta = mg$$

$$T = \frac{mg}{\cos\theta}$$



$$T = \frac{mg}{\sqrt{\ell^2 - \frac{r^2}{4}}}$$

$$T = \frac{\frac{\ell}{mg\ell}}{\sqrt{\ell^2 - \frac{r^2}{4}}}$$

$$T = \frac{\frac{\ell}{mg\ell}}{\sqrt{\ell^2 - r^2/4}}$$

29) $$V = \frac{kq}{z-a} + \frac{k(-q)}{z+a}$$

30)

By Gauss law :

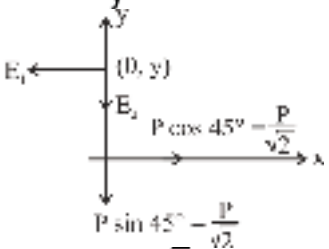
$$\oint \vec{E}.\overrightarrow{dA} = \frac{q_{in}}{\varepsilon_0}$$

If electric field line become discontinuous at Gaussian surface than E = 0

Gauss law become wrong

31) $$E_P = \sqrt{E_1^2 + E_2^2}$$

$$E_1 = \frac{KP/\sqrt{2}}{y^3}$$



$$E_2 = \frac{2KP/\sqrt{2}}{y^3}$$

$$E_P = \frac{KP\sqrt{5}}{\sqrt{2}y^3} = \left(\sqrt{\frac{5}{2}}\right)\frac{kP}{y^3}$$

32)

Work done in moving a charge equals the change in electric potential energy:
$W = q_0 (V_f - V_i)$
Electric potential on the axis of a charged ring:

$$V(x) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{\sqrt{a^2 + x^2}}$$

Potential at the centre (x = 0):

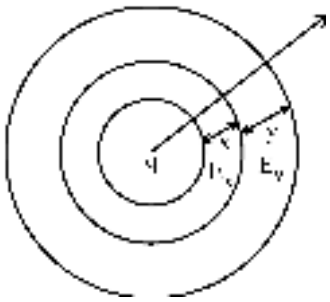
$$V_i = \frac{1}{4\pi\varepsilon_0}\frac{Q}{a}$$

Potential at $x = \sqrt{8}\,a$:

$$V_f = \frac{1}{4\pi\varepsilon_0}\frac{Q}{\sqrt{a^2 + 8a^2}} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{3a}$$

Work done:

$$W = q_0\left(\frac{1}{4\pi\varepsilon_0}\right)\left(\frac{Q}{3a} - \frac{Q}{a}\right) = -\frac{Qq_0}{6\pi\varepsilon_0 a}$$



33)
$\because \quad \Delta V = E.dr$
$E_x \; x = E_y \; y$
$\Rightarrow \quad E_x > E_y$
$\Rightarrow \quad x < y$

34) $\vec{E}$-between plates $= \frac{\sigma}{\varepsilon_0}$

SO, $a = \frac{eE}{m} = \frac{e\sigma}{\varepsilon_0 m}$

$\Rightarrow$ u = 0 so to travel a distance d in time t

$$d = ut + \frac{1}{2}at^2$$

$$d = 0 + \frac{1}{2}\frac{e\sigma}{\varepsilon_0 m}t^2$$

$$\sigma = \frac{2d\varepsilon_0 m}{et^2}$$

35)

τ = P.E. $\sin 30° = 10\sqrt{2}$
P.E = $20\sqrt{2}$ .....(1)
Now potential energy is given by
U = –PE $\cos 30°$

$$U = -\frac{20 \times \sqrt{2} \times \sqrt{3}}{2}$$

= –24.5 J

36)

At null point -

$$\frac{R_{known}}{R_{unknown}} = \frac{\ell_{balanced}}{100-\ell_{balanced}}$$

$$\frac{12}{R_{unknown}} = \frac{60}{100-60}$$

$$\frac{12}{R_{unknown}} = \frac{60}{40}$$

$R_{unknown} = 8\ \Omega$

$$\frac{\rho_{unknown}\ell_{unknown}}{A_{unknown}} = 8$$

$$\frac{\rho_{unknown} \times 1}{0.2 \times 10^{-6}} = 8$$

$\rho_{unknown} = 1.6 \times 10^{-6}\ \Omega.\ m$

37) Initially $\frac{R}{X} = \frac{40}{60} = \frac{2}{3}$

Later $X_{eq} = \frac{10X}{10+X}$

$$\frac{R}{10X}(10+X) = \frac{60}{40} = \frac{3}{2}$$

$$\frac{R}{X}(10+X) = \frac{30}{2}$$

$$\frac{2}{3}(10+X) = 15$$

$$10+X = \frac{45}{2}$$

$X = 22.5 - 10$

$X = 12.5\Omega$

38)

Tension in string T = mg

or $T = F_e$

so $mg = F_e = Q.\frac{\sigma}{2\varepsilon_0}$

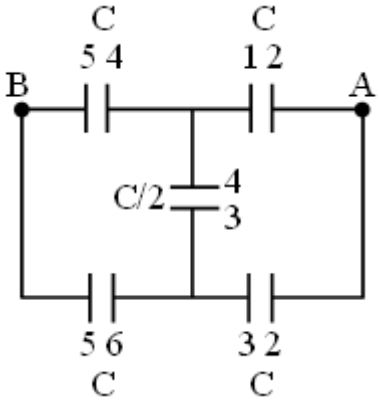
$$mg = Q.\frac{Q}{A2\varepsilon_0}$$

$$Q = \sqrt{2A\varepsilon_0 mg}$$

39) $C_1V_1 = C_2V_2$

$(KC)100 = CV_2$

$V_2 = 100\ K = 100 \times 5 = 500\ V.$



40)

This is balance bridge

$C_{AB} = C = \frac{\varepsilon_0 A}{d} = \alpha \frac{\varepsilon_0 A}{d}$

$\alpha = 1$

41) Let $q_A = q$, $q_B = \frac{8}{8+12} q = \frac{2q}{5}$

$q_C = \frac{12}{8+12} q = \frac{3q}{5}$

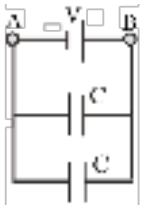
$\therefore q_A : q_B : q_C = q : \frac{2q}{5} : \frac{3q}{5} = 5 : 2 : 3$

42) Net EMF of 9v & 18v cell

$E_{Net} = \frac{18 \times 2 + 9 \times 4}{2+4} = \frac{72}{6} = 12V$

$\therefore v_c = 12 - 9 = 3V$

43) Energy is stored in the space where electric field exits.

44) 

$C_{AB} = 2C$

45) $q = q^0 e^{-t/RC} = CV^0 e^{-t/RC} = 2 \times 5e^{-12/(6\times2)} = \frac{10}{e}$

CHEMISTRY

46) –12250x – 13000 (1 – x) = –12500

750x = 500 $\Rightarrow$ x = 2/3 and y = 1/3

So, $\frac{x}{y} = \frac{2}{1}$

47)

Fact based

48) Theory based

49)

Theory based

50) $K = 0.03\ s^{-1}$ (First order reaction)
$A_0 = 7.2\ mol\ L^{-1}$
$A_t = 0.9\ mol\ L^{-1}$
$7.2 \xrightarrow{t_{1/2}} 3.6 \xrightarrow{t_{1/2}} 1.8 \xrightarrow{t_{1/2}} 0.9$
$t = 3t_{1/2} = \frac{3 \times 0.693}{0.03} = 69.3s$

51)

$\log t_{1/2} = -\log a + C$
$\log t_{1/2} + \log a = C$
$\log t_{1/2} \times a = C$
$t_{1/2} \times a = const$
$t_{1/2} \propto \frac{1}{a}$
Second order

52)

Explain the Question : Find the order of reaction from half-life dependence on concentration.
Concept : This question is based on Half-life relation, Reaction order determination.
Solution : Main rule : For n^{th} order, $t_{1/2} \propto 1/[A]^{n-1}$.

[A] (mol L^{-1})	**$t_{1/2}$(s)**	**Relation**
0.5	100	–
1.0	50	When [A] doubles, $t^{1/2}$ halves

$t_{1/2} \propto 1/[A]$ this matches second order kinetics.
Final answer : Option (3)

53)

NCERT XII, Pg.# 83, Part-I

54)

NCERT Based

55)

Option (3) 0.2 g

56) Theoretical amount (mass) of $NaClO_4$

$$W = \frac{E_{it}}{96500} = \frac{122.5}{2} \times \frac{6.43 \times 10^5}{96500} = 408.12g$$

$$\% \text{ efficiency} = \frac{245}{408.12} \times 100 = 60\%$$

57) $\Lambda_m = \Lambda_m^o - A\sqrt{C}$

Units of $A\sqrt{C}$ = S cm^2 $mole^{-1}$

Uits of A = S cm^2 $mole^{-3/2}$ $L^{1/2}$

58) SRP ↓ Reducing power ↑

59) $$\alpha = \frac{\Lambda_M}{\Lambda_M^\infty} = \frac{3.907}{390.7} = 0.01$$

$[H^+] = C\alpha = 0.01 \times 0.01 = 10^{-4}$

pH = 4

60)

NCERT XII, Part-I, Pg. # 46, 47

61) (a) $Cd(s) + 2Ni(OH)_3(s) \rightarrow CdO(s) + 2Ni(OH)_2(s) + H_2O(l)$

Discharge of secondary Battery

(b) $Zn(Hg) + HgO(s) \rightarrow ZnO(s) + Hg(l)$

(Primary Battery Mercury cell)

(c) $2PbSO_4(s) + 2H_2O(l) \rightarrow Pb(s) + PbO_2(s) + 2H_2SO_4(aq)$

Charging of secondary Battery

(d) $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$ - Fuel cell

62) $$\Lambda_M^o[NH_4OH] = \Lambda_M^o[NH_4Cl] + \frac{1}{2}\Lambda_M^o[Ba(OH)_2] - \frac{1}{2}\Lambda_M^o[BaCl_2]$$

$= 130 + 225 - 120$

$= 10 + 225 = 235$

63) $\lambda_{eq}^{\infty}(BaSO_4) = x_1 + x_2 - x_3$

$\lambda_M^{\infty} = \lambda_{eq}^{\infty} \times V.F. = 2(x_1 + x_2 - x_3)$

$\lambda_M^{\infty} = \frac{k \times 1000}{S}$

$\therefore S = \frac{x \times 1000}{2(x_1 + x_2 - x_3)}$

$BaSO_4$ AB

$K_{sp} = S^2$

$K_{sp} = \frac{2.5 \times 10^{-5} x^2}{(x_1 + x_2 - x_3)^2}$

64) NCERT-XII, Part-I, Pg # 36

65) NCERT-XII, Part-I, Pg. # 43

Given (as written):

$Fe^{2+} \rightarrow Fe^{3+} + e^-$, E° –0.77 V

⇒ reduction potential for $Fe^{3+} + e^- \rightarrow Fe^{2+}$ is

$E^{\circ}_{Fe^{3+}/Fe^{2+}} = +0.77$ V.

$2I^- \rightarrow I_2 + 2e^-$, E° = –0.54 V

⇒ reduction potential for $I_2 + 2e^- \rightarrow 2I^-$ is

$E^{\circ}_{I_2/I^-} = +0.54$ V.

The species with the more positive reduction potential is reduced (cathode):

$Fe^{3+} + e^- \rightarrow Fe^{2+}$ with $E^{\circ}_{cathode} = +0.77$ V.

The other half-cell (I_2/I^-) will act as anode (oxidation),

so use its reduction potential as $E^{\circ}_{anode} = +0.54$ V.

$E^{\circ}_{cell} = E^{\circ}_{cathode} - E^{\circ}_{anode} = 0.77 - 0.54 = 0.23$ V **Final Answer :** Option (3)

66) $E^{o}_{cell} = \frac{0.0591}{n} \log K_{eq}$

$0.295 = \frac{0.0591}{2} \log K_{eq}$

$\log K_{eq} = 10$

$K_{eq} = 10^{10}$

67)

0.1M HCl → $[H^+] = 10^{-1}$

$2H^+ + 2e^- \rightarrow H_2$

$E_{RP} = 0 - \frac{0.06}{2} \log \frac{P_{H_2}}{[H^+]^2}$

$= -0.03 \log \frac{1}{(10^{-1})^2} = -0.06$

100 times dilute ⇒ $[H^+] = 10^{-3}$

$\therefore E_{cell} = -0.03 \log \frac{1}{(10^{-3})^2} = -0.03 \log 10^6 = -0.18$

 E_{cell} decrease by 0.12 V

$$Au^{+} + e^{-} \rightarrow Au(s)\ \Delta G_1^{\circ} \quad \text{—(1)}$$
$$Au^{+3} + 3e^{-} \rightarrow Au(s)\ \Delta G_2^{\circ} \quad \text{—(2)}$$
$$Au^{+} \rightarrow Au^{+3} + 2e^{-}\ \Delta G_3^{\circ} \quad \text{—(3)}$$

68) Eq.(3) = Eq.(1) – Eq.(2)

69)

$r = K_3[N_2O_2][O_2]$

↑

Intermediate

$$\frac{K_1}{K_2} = K_{eq} = \frac{[N_2O_2]}{[NO]^2}$$

$$[N_2O_2] = \frac{K_1}{K_2}[NO]^2$$

$$r = \frac{K_1}{K_2} \times K_3[NO]^2[O_2]$$

70)

Theory based

71)

From I & II experiment
[A] → 2 times
rate → 4 times
Order wrt A = 2
From II & III experiment
[B] → 2 times
r → no change
 order wrt B = 0

72) **Question Explanation**
This question tests your understanding of rate law determination and the difference between molecularity and order of reaction.
Concept
Rate law is determined experimentally, not from the balanced equation. Molecularity refers to the number of species involved in an elementary step, not the overall rate law.
Solution
Statement-1: False
The rate law cannot be predicted just by looking at the balanced chemical equation unless it's an elementary reaction. For most reactions, the rate law is determined experimentally.
Statement-II: False
The sum of powers of concentrations in the rate law is called the order of reaction, not

molecularity.
Molecularity is a theoretical concept applied only to elementary steps and is always a whole number.
Final Answer
Option 3

73)

	2A	→	2B	+	C
t = 0	2atm		0		0
t = 100s	2–2x		2x		x

total pressure = 2–2x + 2x + x

$$2.5 = 2+x$$

$$x = 0.5 \text{ atm}$$

$$k = \frac{2.303}{t}\log\left(\frac{P_0}{P_0 - 2x}\right)$$

$$k = \frac{2.303}{100}\log\left(\frac{2}{2-2(0.5)}\right)$$

$$= \frac{2.303}{100}\log 2$$

$$= 0.693 \times 10^{-2}$$

$$k = 6.93 \times 10^{-3}\ s^{-1}$$

74)

Theoritical

75) $$\frac{\pi = iCRT}{\Delta T_f = KiC} = \pi = \frac{RT \times \Delta T_f}{K}$$

$$= \frac{0.082 \times 298 \times 0.406}{1.86} = 5.34$$

76)

i = 5

$273 - T_f = 5 \times 1.86 \times 0.2$

$T_f = 271.14$ K

$T_f = -1.86°C$

77) Van't Hoff factor 'i' for urea = 1 and for NaCl = 2

$\pi = iCRT$

78)

Theory based

79)

Molarity = 2.05 M → 2.05 moles in 1 L solution.
Mass of solution = 1 × 1.02 × 1000 = 1020 g.
Mass of water = 1020 – (2.05 × 60) = 1020 – 123 = 897 g ≈ 0.897 kg.
Molality = moles ÷ kg solvent = 2.05 ÷ 0.897 ≈ 2.28 mol/kg.
Final answer : 2

80)

$\Delta T_f = i K_f m$
273.15 – 269.28 = i × 1.86 × 1
3.87 = i × 1.86
i = 2 ⇒ n = 2

81)

Relative lowering in vapour pressure ∝ i
(i) 0.1 M urea ⇒ i = 1
(ii) 0.1 M NaCl ⇒ i = 2 (Na^+, Cl^-)
(iii) 0.1 M $CaCl_2$ ⇒ i = 3 (Ca^{+2}, $2Cl^-$)
R·L·V·P is maximum in III
• **Option 1:** Vapour pressure is **highest** in Urea (Solution I) because it has the fewest particles to inhibit evaporation.
• **Option 2:** Relative lowering of vapour pressure is **maximum** in $CaCl_2$ (Solution III) because it produces the highest number of particles.
• **Option 3:** Freezing point is **lowest** for $CaCl_2$ (Solution III) because higher particle counts cause greater freezing point depression.
• **Option 4:** Boiling point is **minimum** for Urea (Solution I) because it has the lowest elevation due to having the fewest particles.
Final answer : Option (**2**)

82)

$P = K_h X$
P ↑ ⇒ Kh ↑ {Same solubility}

83)

$X_B \to 0 \Rightarrow P_S = P^o_A$
$X_B \to 1 \Rightarrow P_S = P^o_B$

84) **<u>Explanation</u>**
• (A) 68% nitric acid and 32% water by mass:
This mixture frms a maximum boiling azeotrope. This is because the strong hydrogen bonding between nitric acid and water leads to a lower vapor pressure than predicted by Raoult's law, causing a higher boiling point. Therefore, (A) matches with (III).
• (B) Solution showing large positive deviation from Raoult's law may form:
Solution with large positive deviations have weaker intermolecular forces than the pure

components. This leads to higher vapor pressures and thus, minimum boiling azeotropes. Therefore, (B) matches with (I).

- **(C) n-hexane and n-heptane:**

These are very similar nonpolar hydrocarbons. Their intermolecular forces are nearly identical, resulting in an ideal solution where Raoult's law is obeyed. Therefore, (C) matches with (II).

- **(D) Two solutions having the same osmotic pressure at a given temperature:**

By definition, solutions with the same osmotic pressure at a given temperature are isotonic. Therefore, (D) matches with (IV).

Correct Matching:

(A) - (III), (B) - (I), (C) - (II), (D) - (IV)

Answer option 4**.**

85)

Theory based

86) **1. For HCI (Heat = x):**

- Equivalents of H^+ = (1 M × 1) × 1 L = 1 eq.
- So, x = (1 eg) x (Constant Heat per equivalent)

2. For H_2SO_4 (Heat = y):

- Equivalents of H^+ = (1 M × 2) × 1 L = 2 eq.
- So, y = (2 eq) x (Constant Heat per equivalent)

Since y is proportional to 2 equivalents and x is proportional to 1 equivalent y = 2x.

Final Answer: Option (4)

87)

St.-I :- 1 mole $\overset{(-2)}{H_2O} \rightarrow \overset{(0)}{O_2}$

V.F of H_2O = 2

g-eq. of H_2O = 1 mol × (V.F = 2)

= 2F

St.-2 :- g-eq. of Al = $\frac{w}{E} = \frac{40}{(27/3)} = 4.44$

= 4.44 F

Final answer : 4

88) $(eq.)_{O_2} = (eq.)_{Cu}$

$$\frac{1.12}{22.4} \times 4 = \frac{w}{63.5} \times 2$$

w = 6.35 g

89) $Ag^+ + e^- \rightarrow Ag$(1 mole)

3 mole → 3 mole Ag

$Cu^{+2} + 2e^- \rightarrow Cu$(1 mole)

3 mole $e^- \rightarrow \frac{3}{2}$ mole Cu

Al^{+3} + 3e → Al (1 mole) will not reduced H_2O will get reduced. {SRP of water is more than Al}

Molar ratio

Ag	:	Cu^{+2}	:	Al^{+3}
3	:	$\frac{3}{2}$	:	0
6	:	3	:	0

90)

$$\log K = \log A - \frac{Ea}{2.3R} \times \frac{1}{T}$$

$$\text{Slope} = \frac{-Ea}{2.3R} = -0.5$$

Ea = 2.3 × R × 0.5

= 2.3 × 2 × 0.5

Ea = 2.3 cal

BIOLOGY

91) NCERT Pg. # 23

92) NCERT Pg.#13,19,20.

93) NCERT Pg.#18,19.

94) NCERT Pg.# 6.

95) NCERT Pg. # 18,19

96) NCERT Page No. 37

97) NCERT Pg.#10-11.

98)

NCERT-XII, Pg # 23 (E), 24 (H)

99)

No. of stamens in flower of solanum ni

100) NCERT Page # 34, 35

101) NCERT Pg. # 72,76,77

102)

NCERT XIth Pg.# 66-67

103)

NCERT XIth Pg.# 65

104)

NCERT XIth Pg.# 64
Chinarose, cotton and lady finger shows twisted aestivation.

105) NCERT (XIth) Pg. # 64(E), 63(H)

106)

NCERT XIth Pg.# 58

107)

NCERT XIth Pg.# 59

108)

NCERT Pg. # 60

109)

NCERT XIth Pg.# 60

110)

NCERT XIth Pg.# 61

111)

NCERT XI, Pg. # 62

112) NCERT XI Pg. 63

113)

NCERT XIth Pg.# 65

114) NCERT Page # 73

115)

NCERT XIth Pg.# 65

116)

NCERT XI Pg. # 89

117) NCERT (XI) Pg. # 89
Cortex, Pericycle and Pith are part of ground tissue system.

118) NCERT XI, Pg.# 90

119)

NCERT XI, Pg. # 24

120)

NCERT XIth Pg.# 76

121) NCERT XI Pg # 92

122) NCERT XI Pg # 76, 77

123) NCERT Pg. # 76

124) NCERT Pg # 77

125) NCERT Pg # 94

126)

NCERt (XI) Pg # 74

127)

NCERT XIth Pg.# 74-76

128) NCERT-XII, Page#20

129)

NCERT-XII, Pg.# 3, 4
Statement-B and Statement-C are correct.

130)

NCERT XIIth Pg.# 5

131)

NCERT XIIth Pg.# 9

132) NCERT-XII, Pg.# 24

133)

NCERT-XII, Pg # 19

134)

NCERT XIIth Pg.# 18

135)

NCERT-XII, Pg. # 19

136)

NCERT -XII Page No. # 30,31

137)

NCERT, Pg. # 33

138)

NCERT Pg.No-36

139) NCERT Pg. # 31,32 (E), 34, 35, 36 (H)

140) NCERT, Pg. # 30

141) NCERT Pg#38,41,42

142) NCERT Pg. # 30

143) NCERT Pg. # 30

144) NCERT Pg. # 31

145) NCERT Pg. # 31

146) NCERT # 37

147) NCERT Pg. 31

148) NCERT Pg. 32

149) NCERT Pg. 32

150)

NCERT, Pg. # 37

151) NCERT Pg. # 32

152) NCERT Pg. # 38

153) NCERT Pg. # 38

154) NCERT Pg. # 26

155) NCERT, Pg. # 35, 36

156) NCERT, Pg. # 37

157) NCERT XII, Pg. # 27,28,31,32

158) NCERT Pg. No. # 53

159) NCERT Pg. No. # 51

160)

NCERT, Pg. # 30

161)

NCERT Pg. # 27

162) NCERT Pg. # 27

163)

NCERT Pg. # 44

164) NCERT(XII) Pg#61/67(H) Para:4.2

165) NCERT-XII, Pg. # 62
NCERT-XII, Pg. # 69

166) NCERT-XII, Pg. # 64, Para-3(E)
NCERT-XII, Pg. # 72, Para-1(H)

167) NCERT (XIIth) Pg. # 60

168)

NCERT, Pg. # 44

169)

NCERT, Pg. # 48

170) NCERT Pg # 58

171) NCERT (XIIth) Pg. # 44, 45

172)

NCERT, Pg. # 48

173)

NCERT, Pg. # 44

174)

NCERT, Pg. # 44

175) NCERT (XII) Pg. # 5, 1, 59, 60, Para-1

176) NCERT-XII, Pg. # 46

177) NCERT Pg. # 48

178)

NCERT, Pg. # 44

179)

NCERT (XIIth) (E)Pg. # 61, Para-3

180)

NCERT, Pg. # 44